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DONATION SAVE LIFE

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APPRECIATION

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Abstract

People lose blood for several reasons, including accidents, surgeries, bleeding, serious injuries, and others. When a patient needs blood transfusion, the hospital contacts blood bank, where blood is collected and stored by donors.

Hospitals cannot be without this important bank that helps save thousands of lives. It also enhances the concept of benevolence and humanity for these donors.

This project aims to employ internet technology in health sector to develop website for blood bank.

The system will facilitate the donating blood operations and speed it up, so the time and effort of donor, needy and banks employees are saved.

The system will be implemented using PHP programming language and MySQL database server, and it allows the donor to register and login to the system, then she can donate blood for the blood bank or to specific blood needy.

الملخص

يفقد الناس الدم لعدة أسباب، بما في ذلك الحوادث والعمليات الجراحية، والنزيف، والإصابات الخطيرة، وغيرها. عندما يحتاج المريض إلى نقل الدم، يتصل المستشفى ببنك الدم، حيث يتم جمع الدم وتخزينه من قبل المتبرعين لا يمكن أن تكون المستشفيات بدون هذا البنك المهم الذي يساعد في إنقاذ الآلاف من الأرواح. كما أنه يعزز مفهوم الخير والإنسانية لهؤلاء المانحين

يهدف هذا المشروع إلى توظيف تكنولوجيا الإنترنت في القطاع الصحي لتطوير موقع إلكتروني لبنك الدم يسهل النظام عمليات التبرع بالدم ويسرعها، وبالتالي فإن وقت وجهد موظفي المتبرعين والمحتاجين والبنوك ينقذون.

سيتم تنفيذ النظام باستخدام لغة برمجة PHP وخادم قاعدة بيانات MySQL ويسمح للمتبرع بالتسجيل وتسجيل الدخول إلى النظام، ثم يمكنها التبرع بالدم لبنك الدم أو لمحتاجين دم معينين.

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TABLE OF ABBREVIATIONS

Abbreviation	Full form	Page number
CI	Confidence Interval	10
SDLC	Software Development Life Cycle	27
ER	Entity-Relationship	40

Chapter one: Introduction

1.1 Introduction

It is a voluntary medical procedure in which blood or one of its compounds is transferred from a healthy person to a sick person who needs blood. This procedure is needed by millions of people every year; It is used during surgery, accidents, or some diseases that require transfusion of some blood components. Importance of donating blood:

Blood transfusion is one of the operations that contribute to saving lives; It is given in such cases:

- 1- When complications occur for pregnant women, such as bleeding before, during or after childbirth.
- 2- For patients during surgical operations such as heart operations, blood vessels, organ transplant surgery and others.
- 3- For people with blood diseases.
- 4- For those injured in accidents.
- 5- Cancer patients.
- 6- Many studies and research have been conducted on the importance of donating blood and organizing initiatives and awareness campaigns about donating blood.

1.2 Background

Despite the growing need for blood in hospitals, blood in banks is not enough due to the small number of donors.

There is a basic need and necessity for the implementation of this project, given the constant need for blood donors and the improvement of the speed of obtaining donation and permanent support for donors. What will make this project special and unique is that it will be an effective website to view the scope of donation to hospitals, in both Arabic and English, and to book a donation appointment and risk questions before approving the donation.

1.3 Objectives

The main objective is to create a website to allow blood donors and blood seekers to obtain blood online.

- 1- Helps facilitate blood donation, Find donors faster.
- 2- Developing a blood management information system to manage blood donor records.
- 3- Awareness of the community about the benefits of donating blood.
- 4- Ease of blood distribution in different hospitals according to hospital requirements.
- 5- Providing safe way collect blood and blood components from volunteer donors around the clock at an affordable cost to the general public and at no cost to the poor.
- 6- Knowing the types of blood types that the bank needs.
- 7- Knowing the person in need of blood by searching for the same blood type within the city

1.4 Problem Definition

The idea of booking a donation to the blood bank came from the lack of this service in Bisha and to many people in need of blood from accidents to births There may be a shortage of blood groups that people need, which are not available in the bank, so they have to look for people who are carrying these blood groups randomly and the condition of the infected person may worsen or die because there is no blood. The study proposed to develop a web site that makes it easier for people to book to donate to the blood bank.

1.5 The importance of the research

Donating blood is a great humanitarian service provided by the citizen to save the lives of others, as one donor can save 3 lives, and therefore the great importance of donating blood, which was allocated by the United Nations on the fourteenth of June of each year to celebrate and raise awareness The importance and benefits of

donating blood are not limited to the recipients only, but also to the donors, as there are many health and psychological benefits for the donors themselves.

1.6 Scope of the project

First, general details about the entity to which the project belongs: Within the Bisha area Scope: King Abdullah Hospital, Maternity Hospital for Women.

1.7 Organizing the research

Is the final form of the search, where the stage of putting the search into its final form is a website where the user books an appointment to donate blood in nearby hospitals in the region, where the site is announced about the types of blood needed by (laboratory blood specialist).

Chapter 2: Dedicated to theoretical studies and we talk about the background of research and topics related to the project and the study of some similar systems.

Chapter 3: Contains the methodology used for the research work and the results of the analysis of the survey.

Chapter 4: Contains project design from database and user interfaces.

Chapter 5: Contains recommendations and future actions.

Chapter 6: System Interface.

Chapter 7: Conclusion and Future Works.

Chapter Two: Literature Reviews

2.1 Background

Provide great humane services to food, to save the lives of others.

Donating blood is a religious duty

Opportunities to donate blood:

When you volunteer to donate blood, you are helping to help people use all blood products. Many patients, meanwhile, donate for the first time to express their gratitude for the recovery of a loved one in their hospital, knowing the importance of donating blood. And understanding your reasons When you donate blood, you will be respected and grateful to the patients who use the blood you donate to save their lives.

The human being can produce 250 million red blood cells every day.

The most important thing in me is in dire need of red blood cells that cannot be built.

Donating blood is very important to save lives when:

- 1- The occurrence of accidents.
- 2- Surgical procedures (e.g., cardiovascular operations).
- 3- Complications for women (such as prenatal bleeding, stoma, etc.).

Donating blood to health institutions.

Blood support provides an emergency blood transfusion service and meets the needs of hospital centers, hospitals, and sanatoriums from blood products for:

Patients with genetic blood diseases, cancerous tumors, renal failure patients, Mediterranean heart patients (thalassemia) and patients with sickle cell anemia (Drepanocytes).

What are the benefits of donating blood for the donor?

Every year around the world celebrates World Blood Donor Day, which falls on June 14. This annual event is an occasion in which verses of thanks are given to voluntary blood donors for free in return for their blood given as a gift to save lives. The level of awareness of the necessity of persevering in donating it to ensure the quality of the quantities and donated products available, and the level of their availability and safety for those patients who need them,

The person becomes able to carry a greater amount of oxygen to the major organs of the body.

And helps to get rid of some iron, which if its level in the blood increases, increases the risk of heart disease, accelerates the oxidation of cholesterol, and increases damage to small arteries.

People who donate their blood at least once a year are less likely to develop circulatory diseases and leukemia.

Donating blood stimulates the marrow cells (the center of blood formation) to replace blood cells with new cells that have a better functional ability and increase the activity of the bone marrow to produce new blood cells (red cells, white blood cells and platelets).

And helping others. Blood donation is a kind of social solidarity that spreads the spirit of love among people.

And there is a reward from God for those who help in the revival of humanity, so you can imagine the amount of reward that God Almighty rewards.

The process of transfusion of blood and blood products helps save millions of lives annually, and can extend the lives of patients with life-threatening conditions and their enjoyment of a better quality of life, and provides support for complex medical and surgical operations. This process also plays an essential role in saving the lives of mothers and children in their care, and during emergency response to man-made and natural disasters. Blood supply services that provide patients with access to safe blood and blood products in sufficient quantities are a key component of an

effective health system. It is only possible to ensure an adequate supply of blood thanks to the regular donation of blood by voluntary and unpaid donors. However, the existing blood supply services in many countries face the challenge of providing enough blood and ensuring its quality and safety at the same time.

What happens to the donated blood?

The donated blood is divided into red blood cells, platelets and plasma, and then stored safely until needed in a medical procedure.

Risks of donating blood:

Donating blood does not come with any risks. It usually does not pose any danger to the human body, and the materials used are all sterile and have only use, and diseases cannot be transmitted through the donation process. Especially if the safety instructions related to the method of blood withdrawal and sterilization are followed, so all medical institutions use new needles to withdraw the blood sample from each person, and in some cases the person may suffer from some symptoms caused by donating blood such as nausea, dizziness, feeling the possibility of fainting, and it will disappear. These symptoms usually do not exceed 3 days [1].

2.2 Previous Studies

2.2.1 Knowledge about blood donation among a sample of Thai university students:

V. Wiwanitkit, MD, (2002) submit a study of knowledge about blood donation among a sample of Thai university students use A self-administered questionnaire and face-to-face interview.

The results: Although most participants (80%) knew about blood donation, only 11% (44 subjects) had ever donated blood voluntarily. The study found no significant correlation between demographic data (gender, age, educational level of subjects studied) and such knowledge or actual blood donation. Among the non-donor respondents, fear (305 cases) was the most common reason for not donating

blood [2]. What distinguishes this study is its assessment of voluntary unpaid blood donation donors.

2.2.2 The blood-donation experience: perceived physical, psychological, and social impact of blood donation on the donor:

Birgitta Nilsson Sojka (2003) submit a study on the blood-donation experience: perceived physical, psychological and social impact of blood donation on the donor Use A self-administered questionnaire.

The results Altogether, 54% (287 out of 528) of the blood donors reported one or several effects.

Exclusively positive effects were described by 29% (151) of blood donors, while exclusively negative effects and mixed effects (i.e., concomitant positive and negative effects in the same subject) were described by 19% (103) and 6% (33), respectively, while no effect was reported by 46% (241) of the donors. Among positive effects a feeling of satisfaction, of being more alert, and feeling generally better than before the blood donation predominated for both female and male donors. Among negative effects, vertigo/dizziness was reported more frequently by female donors ($P < 0.0001$). [3] the open question in the questionnaire did not give a clear result, unlike if it was a specific question. Current study focuses on closed questions.

2.2.3 Knowledge, attitudes and practices regarding blood donation among the Saudi population:

Maqbool Allam, Bourhan El Din Masalmeh (2004) a study on Knowledge, attitudes and practices regarding blood donation among the Saudi population use interview and a questionnaire in Arabic language. The results Out of 500 individuals, 291 (58.2%) were donors and 209 (41.8%) were non donors. Four hundred and seventy-one (94.2%) replied that a person more than 45-years of age could not donate blood. Fifty-one-point two percent (107 out of 209) of the non-donors answered that one can donate blood once a year. Out of 291 donors, 186 (63.9%) donated blood to their family members or friends and 105 (36.1%) were volunteer blood donors.

Eighty-nine non donors (42.6%) replied that they were not approached by anybody for blood donation and 80 (38.3%) considered themselves 'unfit' for donation due to weakness.[4] What distinguishes this study is the accuracy of the result. The disadvantage of this study is that it was limited to males only. The current study focuses on males and females in general.

2.2.4 The blood donation experience: self-reported motives and obstacles for donating blood:

Birgitta Nilsson Sojka (2007) Submit a study on the blood donation experience: self-reported motives and obstacles for donating blood use self-administered questionnaire

The results: Altogether 531 whole blood donors filled in the questionnaire (88.5%; 322 men and 209 women). No statistically significant differences were found between male and female blood donors concerning general reasons and motives related to donating blood. The most frequently reported reasons for giving blood the first time were 'influence from a friend' (47.2% of donors) and 'request via media' (23.5% of donors). Among general reasons/motives with highest ranking of importance, the most reported motive for donating blood were 'general altruism' (40.3%), 'social responsibility/obligation' (19.7%) and 'influence from friends' (17.9%). 'General altruism' and 'social responsibility/obligation' were also the most frequent reasons for continuing to donate blood (68.4 and 16.0%, respectively). The most commonly reported obstacle to becoming a regular blood donor was 'laziness' (19.1%) followed by 'fear of needles' (10.5%).[5]

The advantage of this study is to focus on influencing people to donate blood.

2.2.5 Complications related to blood donation: a population-based study:

B. S. Sorensen (2007) Submit a study on Complications related to blood donation: a population-based study uses Data from a survey conducted in 2003 in Aarhus County, Denmark, were used to assess the overall rate of donor complications.

The results: In the regional survey, we identified 340 complications of any type among 41 274 donations, corresponding to a rate of 824/100 000 donations [95%

confidence interval (CI): 741–916]. All complications were either needle injuries or vasovagal reactions. In the nationwide register, a total of 752 moderate and severe complications were recorded among 2 575 264 donations, corresponding to a rate of 29/100 000 donations (95% CI: 27–31). The rates of complications leading to long-term morbidity or disablement ($> 5\%$ loss of working capacity) were 5/100 000 donations (95% CI: 4·2–5·9) and 2·3/100 000 donations (95% CI: 1·8–2·9), respectively.[6] The study focused on complications related to blood donation.

2.2.6 Blood Donation Prediction using Artificial Neural Network:

Alaa M. Barhoom (2019) Submit a study of Blood Donation Prediction using Artificial Neural Network use questionnaire. Their results: it was found that the ANN model using JustNN tool led to the best test set performance accuracy of (99.31%).[7] The use of the artificial neural network gave an accurate result.

2.3 Related Applications

2.3.1 Wateen

Figure 2.1 Wateen App It is the official blood donation application in Saudi Arabia, whether you are a blood donor, not eligible to donate, or a representative of a blood bank. [8]

Application Features:

- 1- Request a blood donation from Wateen community and share donation requests on social networks
- 2- List and websites of blood banks
- 3- Follow up on donation history
- 4- Reminders of donation dates and donation campaigns around you

Disadvantages:

- 1- The large number of users reduces the efficiency and quality of the program.



Figure 2.1 Wateen App

2.3.2 Red Blood donation:

Figure 2.2 Red App Every time you donate blood, 4 different people get a new lease on life, red is the first application in Egypt that facilitates the process of donating blood by connecting donors to people who need blood.[9]

Application Features:

- 1- Identify the location of the closest person in need of blood in your surroundings of the same blood type
- 2- Communicate with the patient to whom you have donated yourself and make sure that blood has reached him and check on him after donating.
- 3- If the donor cannot donate blood, he can share the needy condition with someone or he can invite a friend to donate through the application.
- 4- Every time you donate blood from Application Red
- 5- You will take points that you can exchange for a discount or subscriptions from some platforms.

Disadvantages:

- 1- The person who needs blood is the one who enters his data in the application, so he can wait a long time for the donor person.
- 2- The registration page in the program is slow to load.

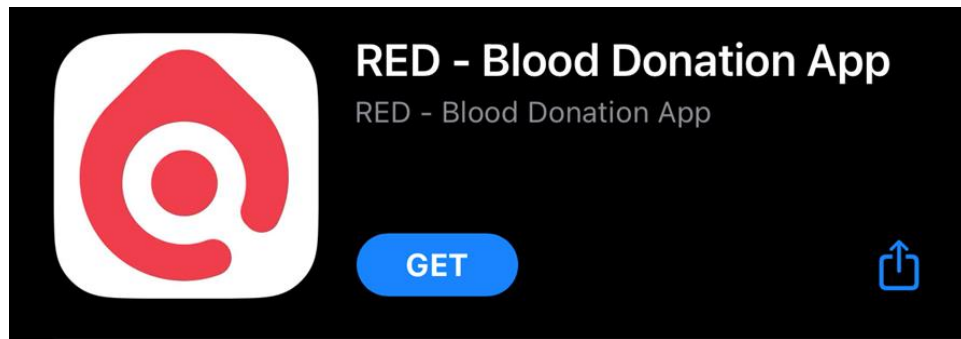


Figure 2.2 Red App

2.3.3 A comparative study:

Table 2.1 A comparative study

System	Advantages	Disadvantages
Wateen	- Request a blood donation from Wateen community and share donation requests on social networks - List and websites of blood banks - Follow up on donation history - Reminders of donation dates and donation campaigns around you	The large number of users reduces the efficiency and quality of the program
Red Blood donation	- Identify the location of the closest person in need of blood in your surroundings of the same blood type - Communicate with the patient to whom you have donated yourself and make sure that blood has reached him and check on him after donating. - If the donor cannot donate blood, he can share the needy condition	The registration page in the program is slow to load.

2.4 Summary:

In this chapter, previous studies on blood donation are discussed in detail.

Chapter Three: Project Methodology

3.1 Introduction:

Development of any system needs to follow one of the software development methodologies, as it helps in ordering the software development steps and help completing the tasks. In this chapter we are going to show the methodology of our proposed system.

3.2 Project work stages:

Figure 3.1 Project work stages Following the proposed methodology will help in managing the proposed system. The system's waterfall development methodology was used in current study. The following figure 4 show the waterfall methodology.

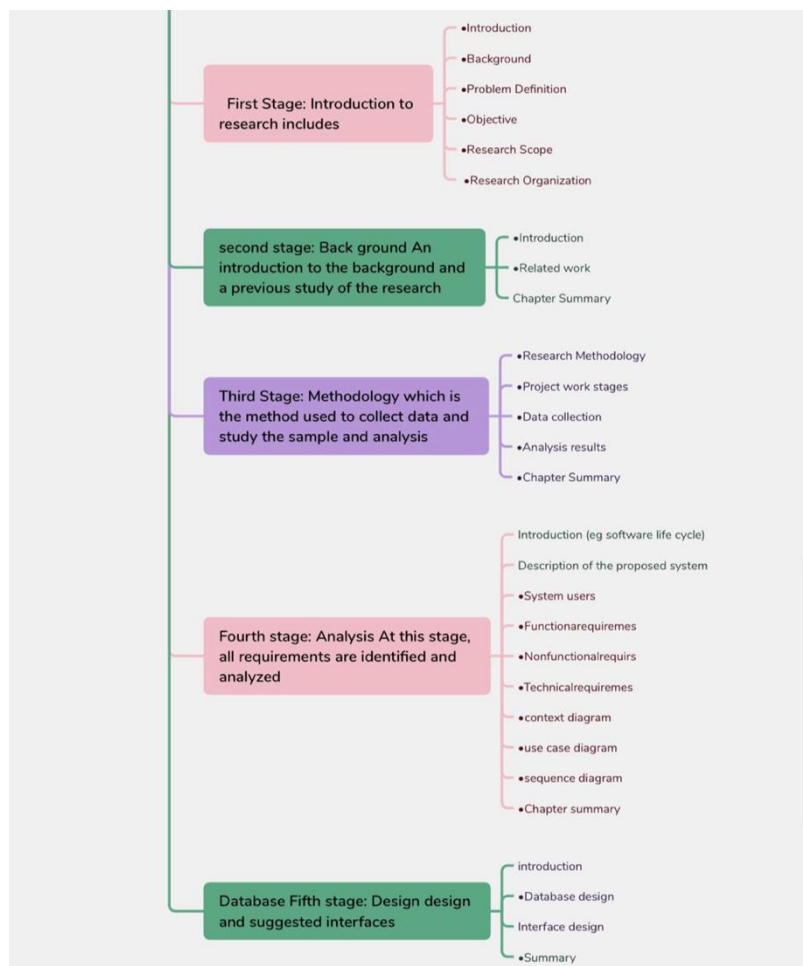


Figure 3.1 Project work stages

3.3 Development Life Cycle:

Waterfall model maintains that moving from one phase to another happens only when its preceding phase is reviewed and verified. Time spent early in the software production cycle can reduce costs at later stages, in common practice, waterfall methodologies result in a project schedule with 20–40% of the time invested for the first two phases, 30–40% of the time to coding, and the rest dedicated to testing and implementation

The waterfall methodology steps are:

- 1- Initiating: In this step an idea was proposed and presented to the supervisor to accept the idea.
- 2- Project planning: In this step, teamwork was done to develop project steps, determine which software to be used, hardware and tools.
- 3- Analysis: which result in determines the project requirements and scope in the first phase therefore, this is appropriate for the project, as the requirements are clear, and the scope is well defined, so the cost of change is low.
- 4- Design: In this step the system database, interface and objects that will consist of the project are designed.
- 5- Implementation: is the process by which the actual system is implemented using the proposed framework and making the system actual.
- 6- Testing: In this step the test is deployed for each project unit and the entire system is combined as a single unit.
- 7- System delivery: At this final stage, the actual system is produced and applied to the project scale.

3.4 Timeline (Gannet chart):

The timeline shows what periods to be spent building the project and according to each stage of system development, namely requirements analysis design, implementation, verification, and maintenance.

Table 3.1 Project timeline

Task Name	Duration	Start	End
Examination	125 Days	9/08/2022	1/02/2023
Project Initiating	14 Days	9/08/2022	9/18/2022
Project Planing	14 Days	9/08/2022	9/20/2022
Project Analyses	20 Days	9/22/2022	10/10/2022
Project Design	14 Days	10/13/2022	10/27/2022
Project implementation	30 Days	11/01/2022	11/29/2022
Project Testing	14 Days	12/01/2022	12/13/2022
Project Delivery	0 Days	1/02/2023	1/02/2023

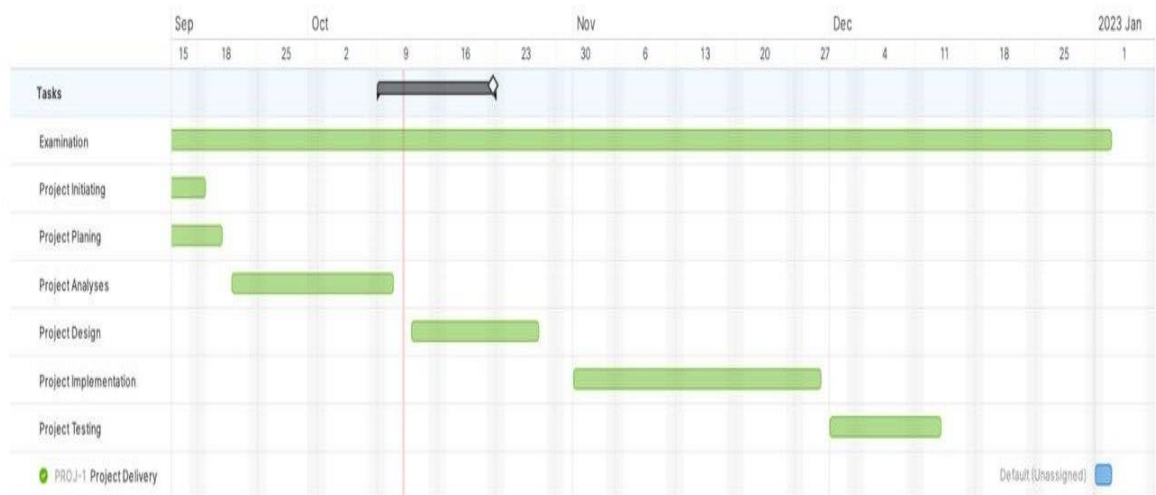


Figure 3.2 Gannet chart

3.5 Data Collection:

The concept of the sample: the sample can be defined as a subset of the study population that is selected in an appropriate manner, conducting the study on it, and then using those results and circulating them to the original study population as a whole. All units of society.

All data related to the research, which contributed to identifying the problems, was collected by creating a query under the name (WAREED to donate blood) on the Google Drive platform and publish it to the community and residents of the Bisha area through this link.

3.6 Sample collection and study:

Figure 3.3 Questionnaire After entering the survey link by clicking on it, this main screen for the survey will appear asking you to answer the questions by selecting the answer from the options presented. The following is a questionnaire.



Figure 3.3 Questionnaire

3.6.1 Questionnaire Questions:

Q1: Your sex?

- a. Male
- b. Female

Q2: Do you know your blood type?

- c. Yes
- d. No

Q3: Have you ever donated blood?

- e. Yes
- f. No

Q4: If you have donated blood before, how was your experience?

- g. Excellent
- h. Very well
- i. bad
- j. I did not donate

Q5: Would you like to donate blood but have some problems?

- k. Yes
- l. No

Q6: Do you think having an electronic method will make it easier for you to donate?

- m. Yes
- n. No

Q7: What is your blood type?

- o. A+
- p. B+
- q. O+
- r. AB+
- s. A-
- t. B-
- u. O-
- v. AB-
- w. I don't know.

Q8: What are the reasons for your donation?

- x. Scarcity of your blood type
- y. Patient Assistance
- z. Renewal of your blood cells
- aa. Charity work "Charity on your health"

Q9: Do you inspect periodically on an ongoing basis?

bb. Yes

cc. No

3.6.2 Questionnaire analysis results:

The number of despondences reached 316 responses on the online created questionnaire. The collected responses were analyzed, and the results were listed as follows:

Figure 3.4 Gender This question was about the gender of the survey participants, and it turned out that 92% of the participants were female and 7.9% of the participants were male.

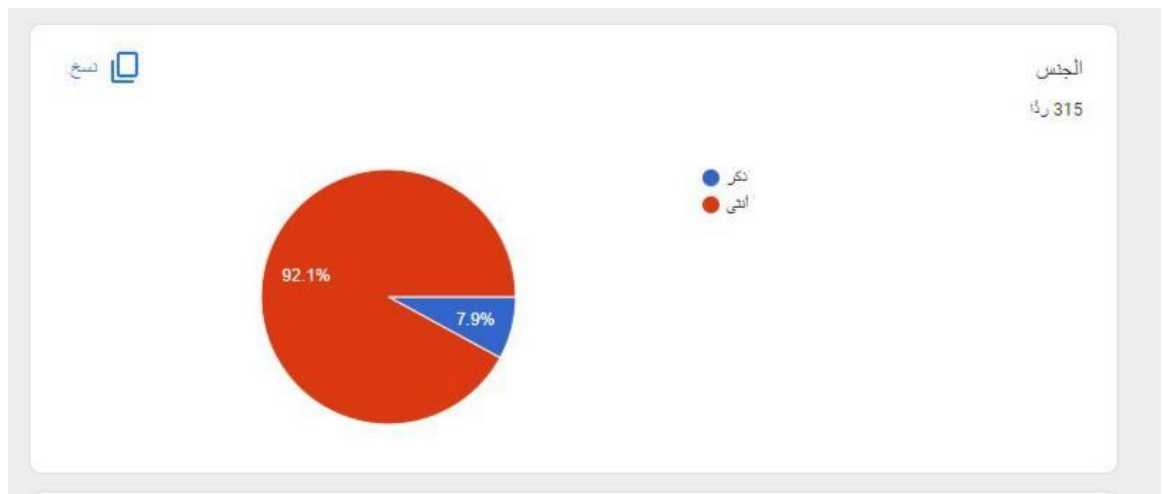


Figure 3.4 Gender

Figure 3.5 History of blood donation This question is about the date of donation and it turns out that 85% have not made a donation before and 14.6% have made a donation before.

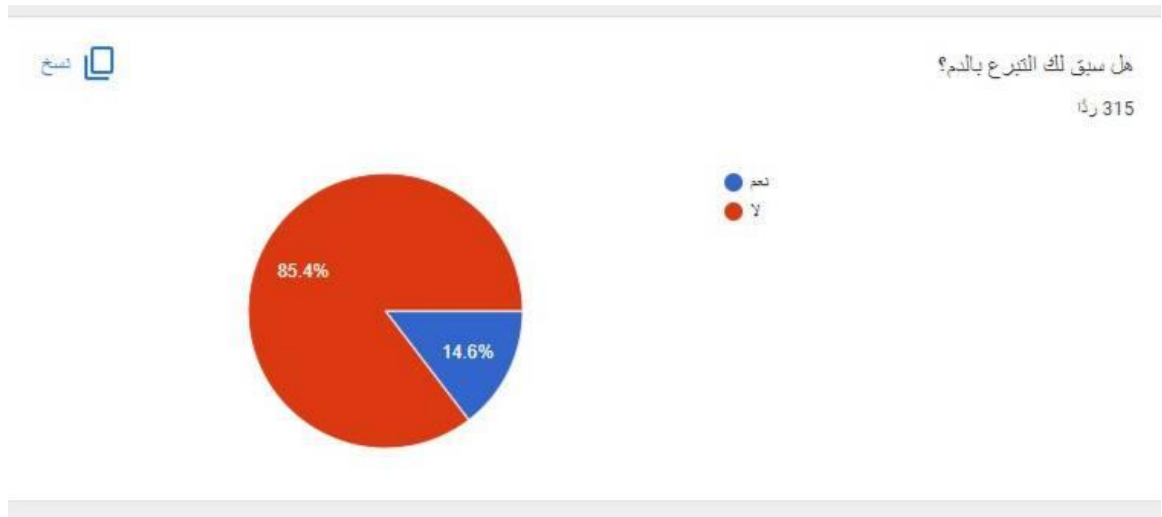


Figure 3.5 History of blood donation

Figure 3.6 Blood donation experience This question about the blood donation experience shows that 84.4% have never donated before, 12.7% of their experience was excellent and 2.9% had a very good experience.



Figure 3.6 Blood donation experience

Figure 3.7 Knowing the blood type This question about knowing the blood type shows that 66% of the participants know their blood type and 34% do not know it.

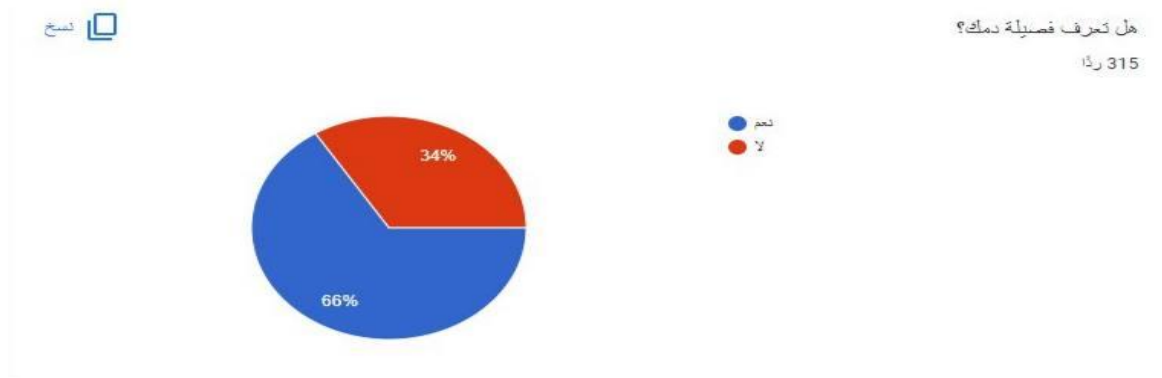


Figure 3.7 Knowing the blood type

Figure 3.8 Desire to donate This question about the desire to donate blood shows that 64.8% have a desire to do so and 35.2% have no desire.

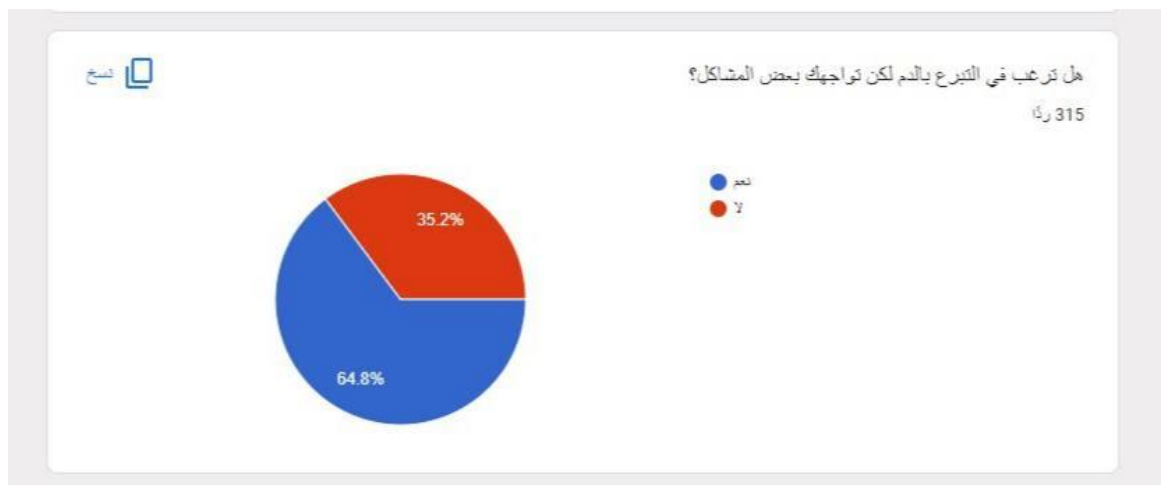


Figure 3.8 Desire to donate

Figure 3.9 Presence and electronic entertainment to donate This question about the existence of an electronic means that facilitates blood donation shows that 91.7% support its existence and 8.3% do not support it.

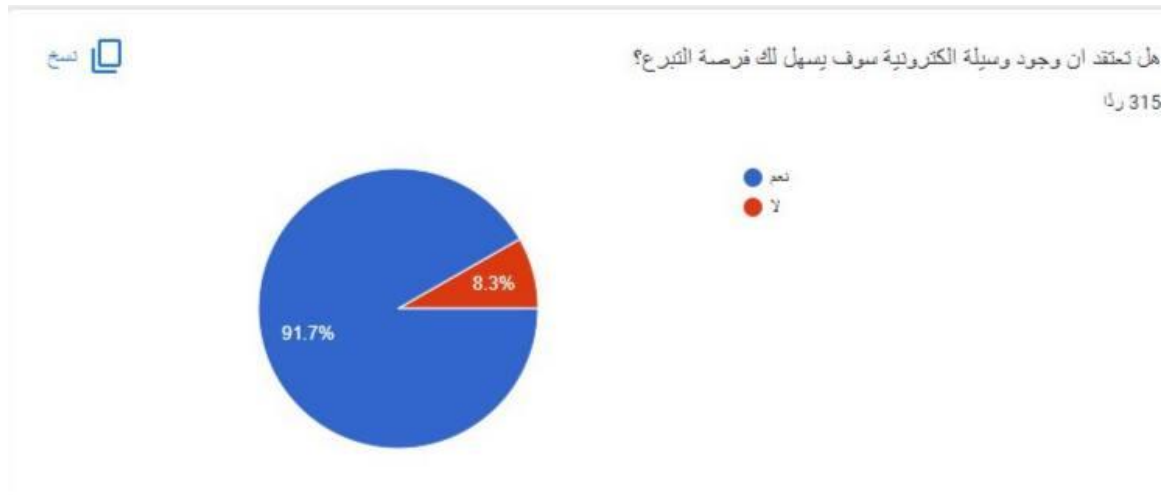


Figure 3.9 Presence and electronic entertainment to donate

Figure 3.10 Blood type This question about blood type shows that most of the participants in the questionnaire carry B+ blood type.

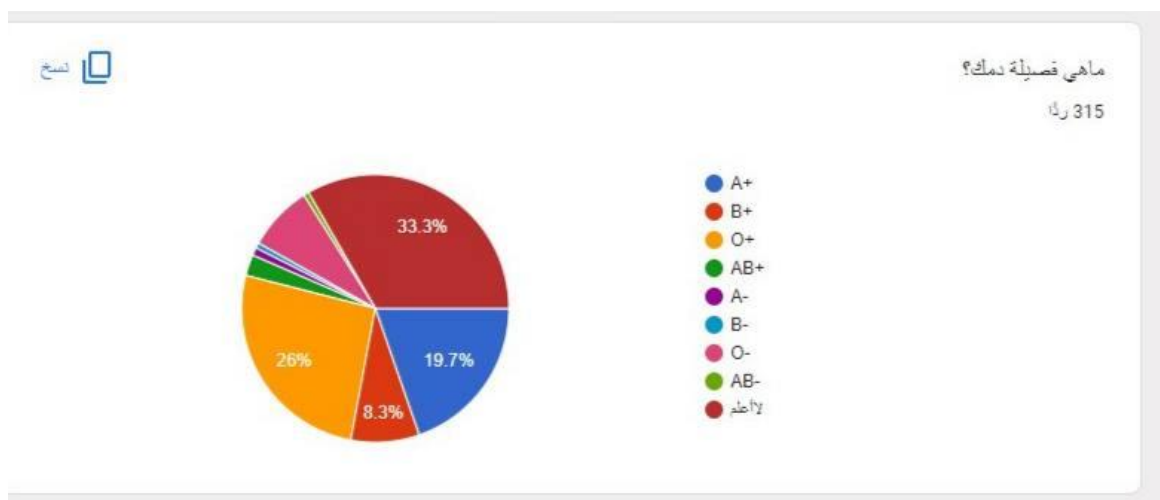


Figure 3.10 Blood type

Figure 3.11 Reasons for donation This question about the reasons for donating shows that 62.2% want to donate to a charity, 28.3% to help a patient and 6% for the scarcity of blood type.

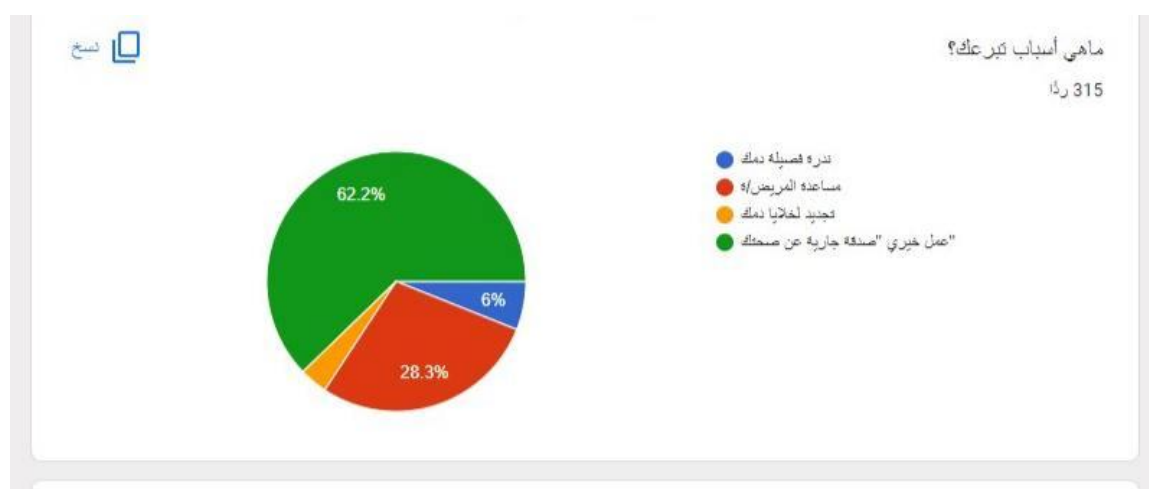


Figure 3.11 Reasons for donation

Figure 3.12 Periodic examination This question about periodic examination shows that 76.5% did not do the periodic examination and 23.5% did.

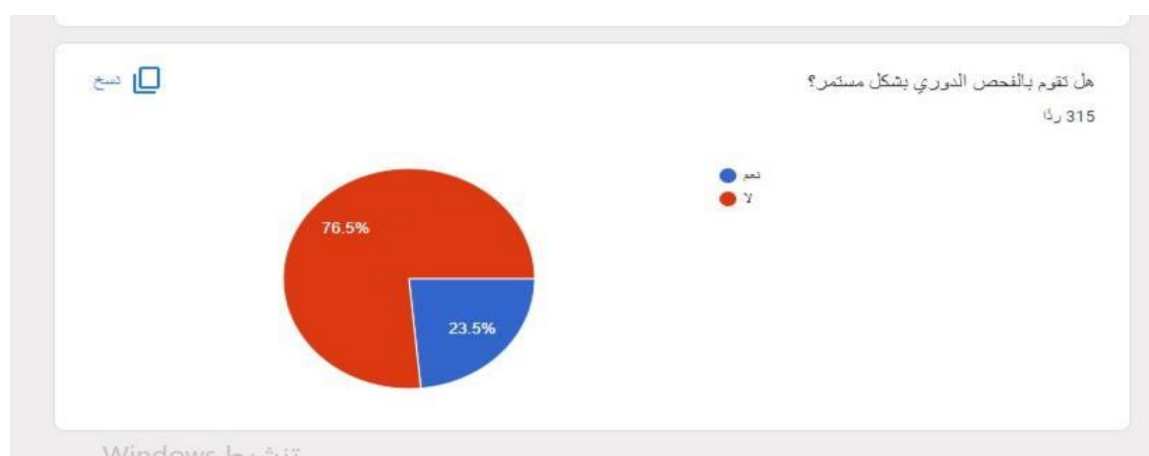


Figure 3.13 Periodic examination

3.7 Result and Discussion:

After collecting the data shown, some of the basic information of the project in the Bisha area, it was found that there is a large number of women who need to donate blood at a rate of 92.1%, and there are people who do not know their blood type at a rate of 66.1% and confirmed from the community that they have a prior donation of 85.4% It is a very high percentage and also here many people face problems when donating blood with a rate of 64.9% and there is a very great belief that the presence of the website will facilitate donation operations

in the region by 91.1% and this is a good motivator and that there are people who confirmed that their blood donation was an ongoing charity on their health, and that there are people who do a continuous blood test with a rate of 76.5%.

3.8 Summary:

In this chapter, the research methodology, the stages through which the project will be implemented, the time plan for the project's workflow, and the data collection mechanism attached to the results were discussed. The next chapter will address the system analysis process.

Chapter Four: System Analysis

4.1 Introduction:

Figure 4.1 SDLC The software life cycle (SDLC) is the set of stages and phases that any software system goes through from its beginning to its end, SDLC consists of several clearly defined and differentiated work phases that systems engineers and systems developers use to plan, design, build, test, and deliver information systems. SDLC is a way to measure and improve the development process, and allows for accurate analysis of each step of the process, which in turn helps companies maximize efficiency at each stage, while increasing computing power. SDLC simply defines each task required to create a software application. This helps reduce waste and increase the efficiency of the development process, also ensures that the project stays on correct track [10].



Figure 4.1 SDLC

4.2 Description Of the Proposed System:

All aspects of the proposed system will now be described.

4.2.1 Proposed system users:

- 1- Administrator of the website: this user can manage users and operations.
- 2- Blood donation officer: this user can manage the needs for the blood donation in all hospitals in Bisha city.
- 3- Donor: this user is the person who wants to donate blood.

4.2.2 Functional Requirements:

The system provides each user with many functions and services, these services and functions known as functional requirements [11], the functional requirements for the proposed system are:

- **Administrator must be able to:**
 - 1- Login.
 - 2- Manage registered blood donation officials.
 - 3- Manage registered donors.
 - 4- Change password.
 - 5- Update profile.
 - 6- Logout.
- **Blood donation officer must be able to:**
 - 1- Create account.
 - 2- Login.
 - 3- Manage blood needs.
 - 4- Manage working hours.
 - 5- View reservations.
 - 6- Change password.
 - 7- Update profile.
 - 8- Logout.
- **Donor must be able to:**
 - 1- Create account.
 - 2- Login.
 - 3- Checking the validity of donating blood.
 - 4- Reserve donation appointment.
 - 5- Cancel reservation.
 - 6- View blood needs.
 - 7- Change password.
 - 8- Update profile.
 - 9- Logout.

4.2.3 Non-Functional Requirements:

Non-functional requirements describe how the system works and what are the properties of the system [12].

Table 4.1 Non-functional requirements

Requirements	Details
Usability	The user interface of the proposed website will be designed to be easy to use and easy to understand by all users.
Reliability	The proposed website will work fine for long time and should not be affected by the number of users by using powerful database server (MySQL).
Performance	The proposed website will respond to the users' request very fast by writing good and efficient code for the application.
Availability	The proposed website will be available all the time.
Security	The proposed website will provide a high security level to prevent anyone who didn't have permissions from changing the system data or getting users' information and allowing each user to login using username and password.
Extensibility	The proposed website will be ready for adding new features in the future like Arabic language and SMS notifications and other.
Documented	The proposed website will be fully documented specially the code and the report of the graduation project will have full description about the website.

4.2.4 Technical Requirements:

- **Hardware Requirements**

- 1- Laptop with core i5, 4GB Ram.
- 2- Internet connection 4G/ ADSL.

- **Software Requirement**

- 1- PHP: is a server-side scripting language designed for web development but also used as a general-purpose programming language.
- 2- HTML: The Hypertext Markup Language used to build web interfaces.
- 3- CSS: Cascading Style Sheets (CSS) is a style sheet language used for describing the look and formatting of a document written in a markup language.
- 4- Notepad++: Text editor for writing PHP codes.
- 5- MySQL: is the database server that is used to create database of the system.
- 6- Photoshop: photo editor.

4.2.5 Context Diagram:

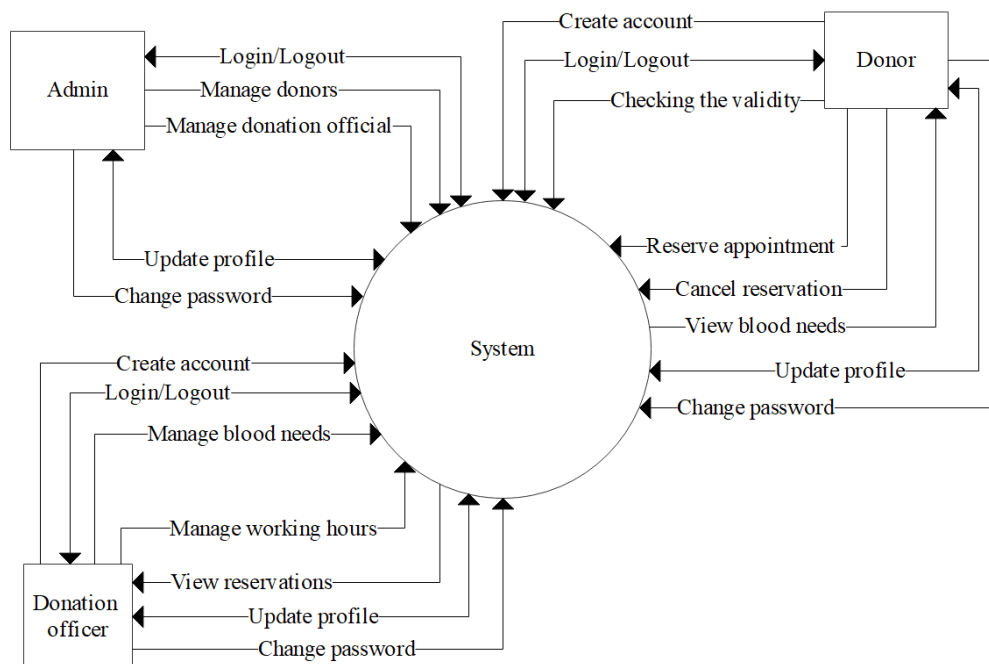


Figure 4.2 Context diagram

4.2.6 Use Case Diagram:

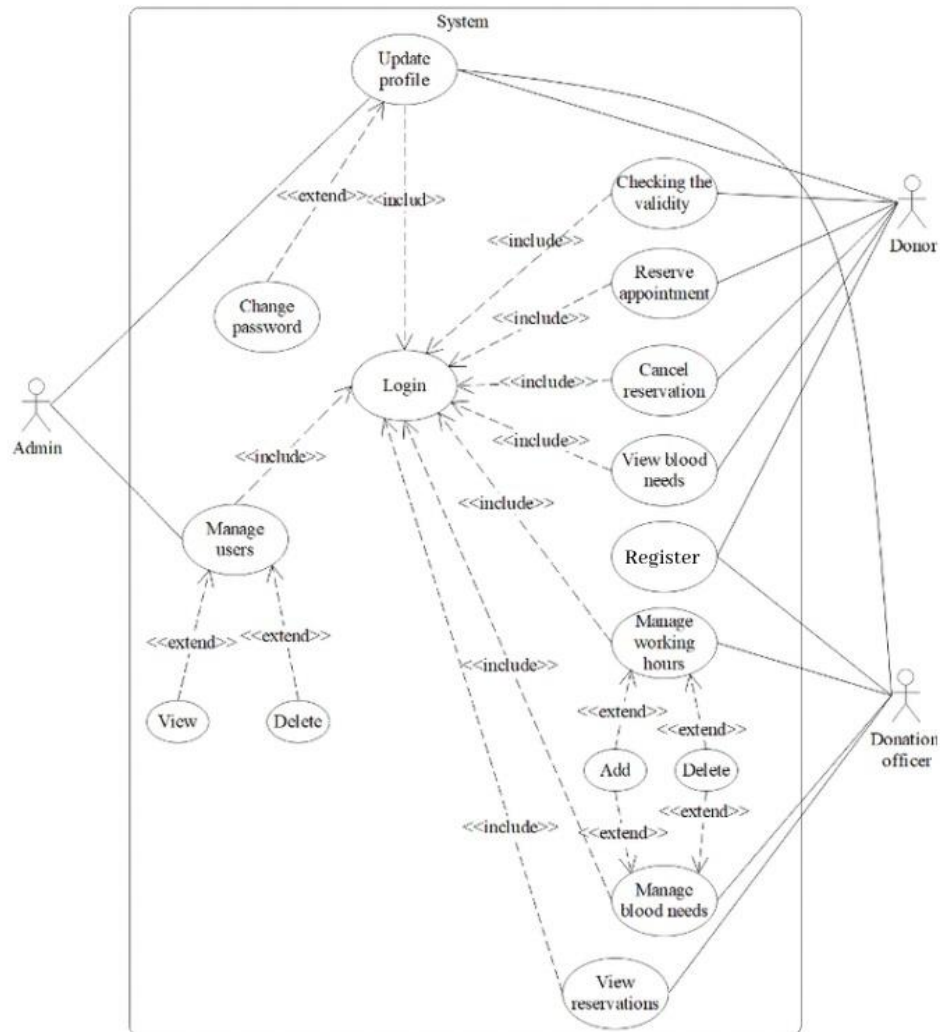


Figure 4.3 Use case diagram

4.2.7 Sequence Diagrams:

Figure 4.4 shows the register sequence diagram, in this diagram the messages and replies between system components in register account are shown.

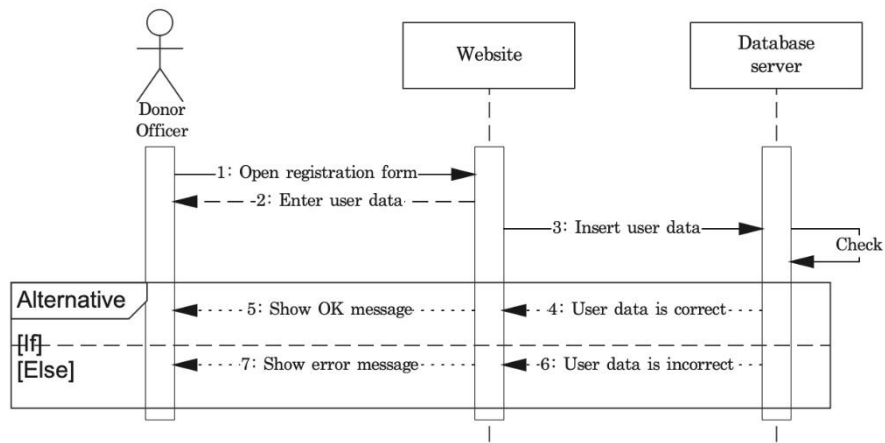


Figure 4.4 Register sequence diagram

Figure 4.5 shows the login sequence diagram, in this diagram the messages and replies between system components in login are shown.

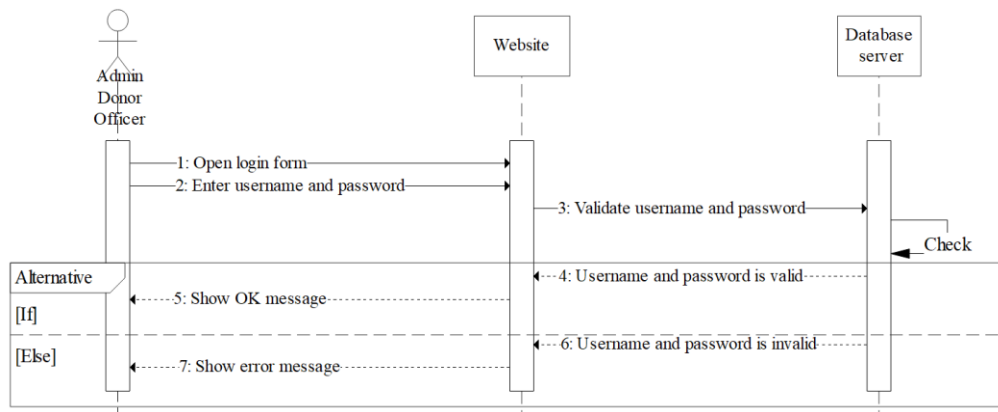


Figure 4.5 Login sequence diagram

Figure 4.6 shows the update account sequence diagram, in this diagram the messages and replies between system components in update account are shown.

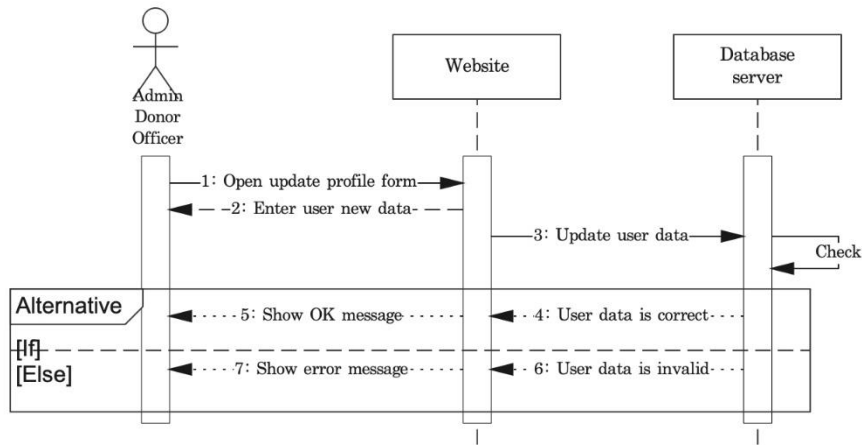


Figure 4.6 Update account sequence diagram

Figure 4.7 shows the change password account sequence diagram, in this diagram the messages and replies between system components in change password are shown.

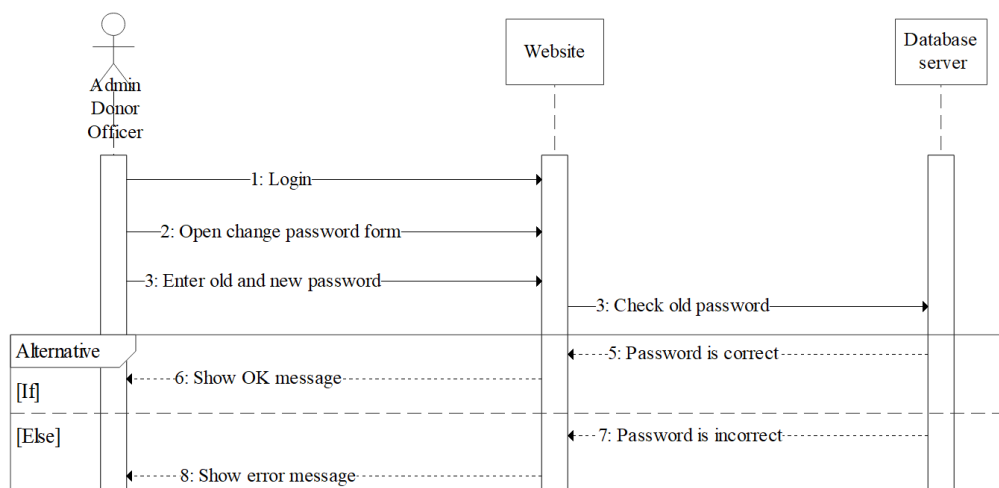


Figure 4.7 Change password sequence diagram

Figure 4.8 shows the delete officer account sequence diagram, in this diagram the messages and replies between system components in delete officer are shown.

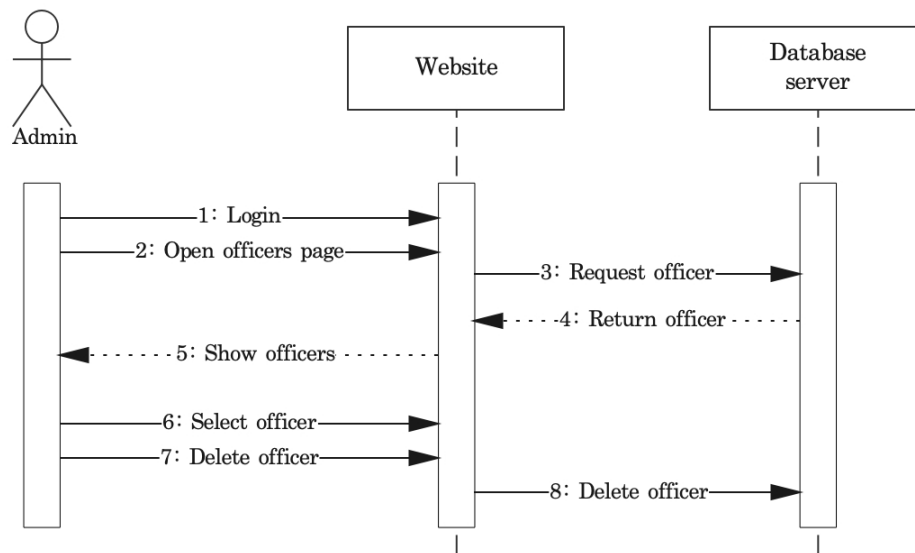


Figure 4.8 Delete officer sequence diagram

Figure 4.9 shows the delete donor account sequence diagram, in this diagram the messages and replies between system components in delete donor are shown.

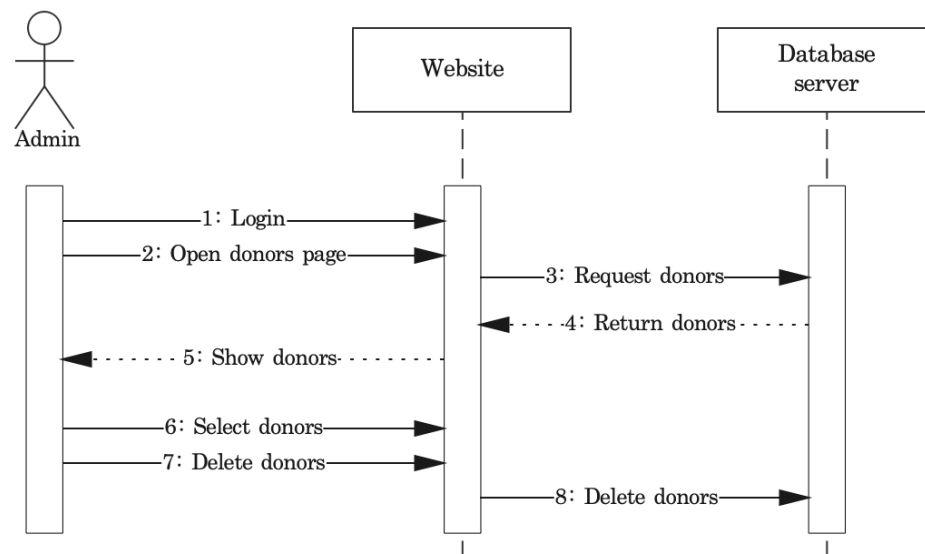


Figure 4.9 Delete donor sequence diagram

Figure 4.10 shows the add need sequence diagram, in this diagram the messages and replies between system components in add need are shown

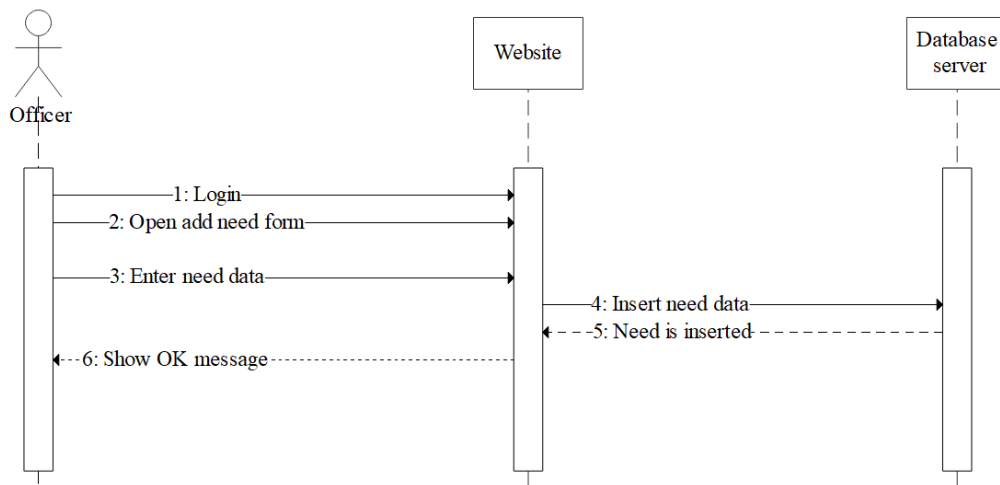


Figure 4.10 Add need sequence diagram

Figure 4.11 shows the delete need sequence diagram, in this diagram the messages and replies between system components in delete need are shown

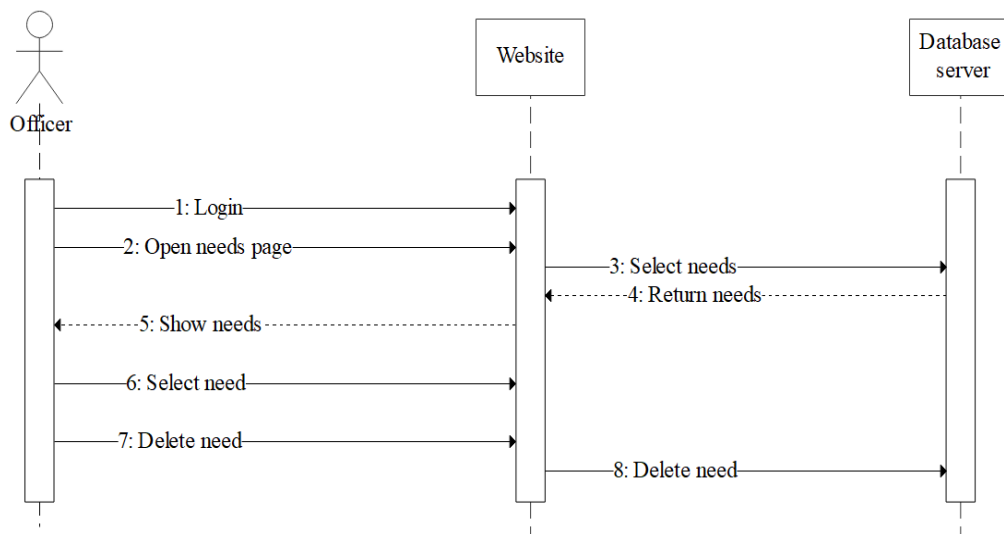


Figure 4.11 Delete need sequence diagram

Figure 4.12 shows the add working hour sequence diagram, in this diagram the messages and replies between system components in add working hour are shown.

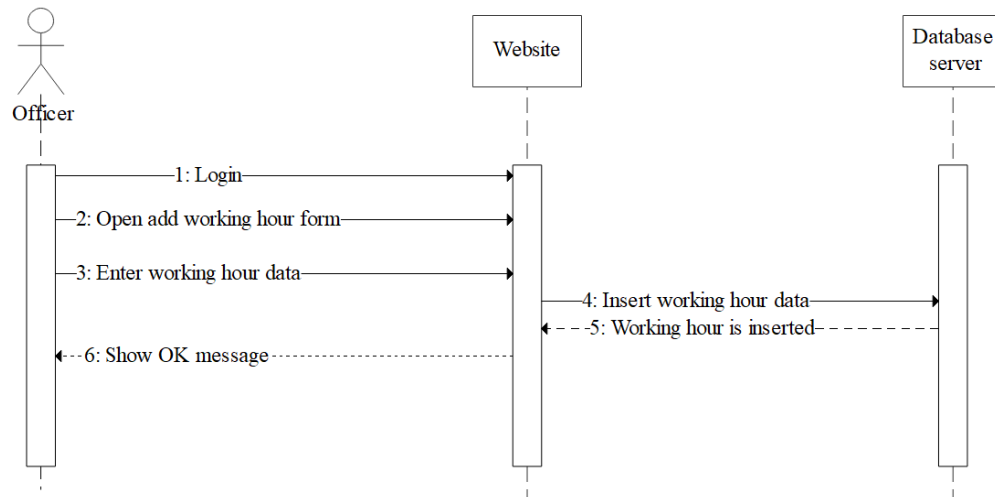


Figure 4.12 Add working hour sequence diagram

Figure 4.13 shows the delete working hour sequence diagram, in this diagram the messages and replies between system components in delete working hour are shown.

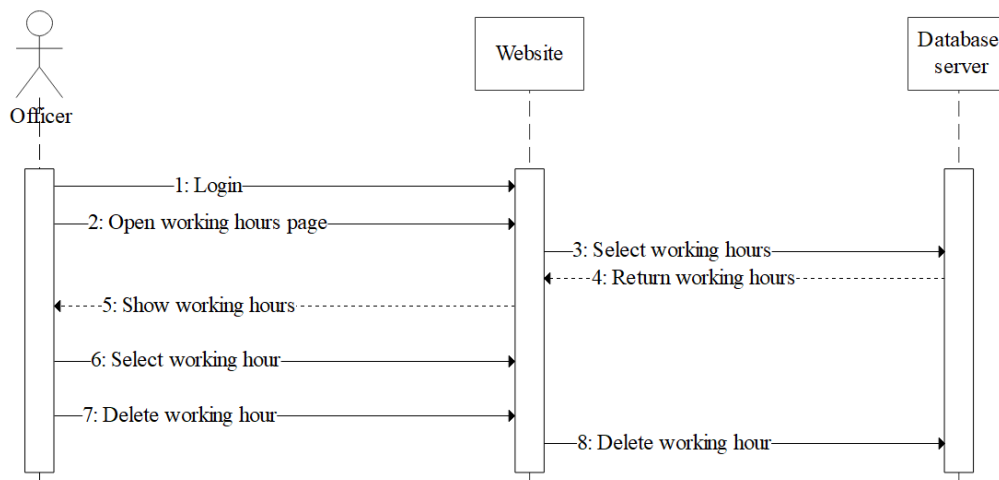


Figure 4.13 Delete working hour sequence diagram

Figure 4.14 shows the view blood need sequence diagram, in this diagram the messages and replies between system components in view blood need are shown.

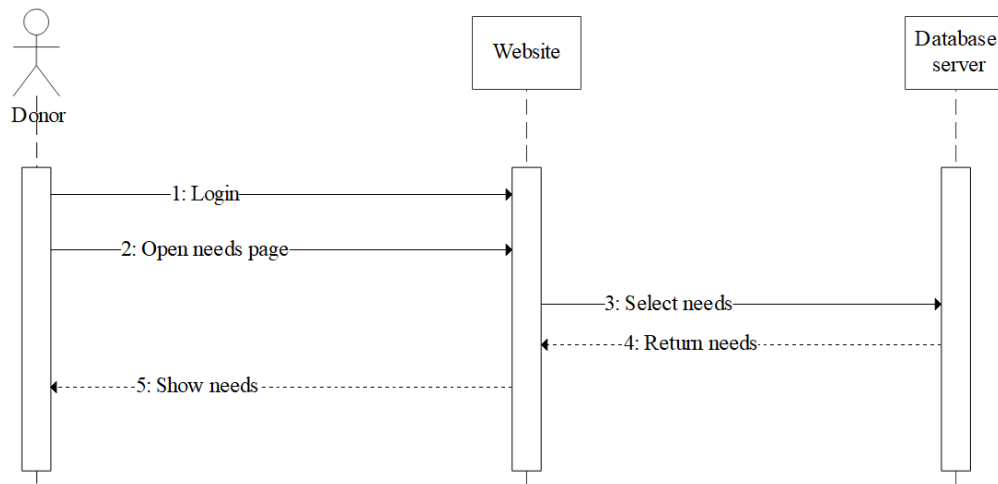


Figure 4.14 View blood need sequence diagram

Figure 4.15 shows the reserve appointment sequence diagram, in this diagram the messages and replies between system components in reserve appointment are shown.

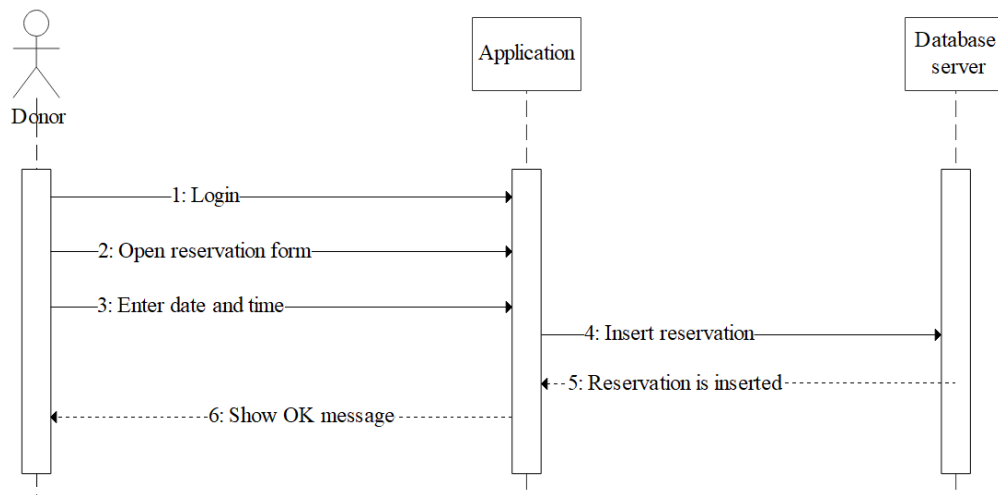


Figure 4.15 Reserve appointment sequence diagram

Figure 4.16 shows the cancel reservation sequence diagram, in this diagram the messages and replies between system components in cancel reservation are shown

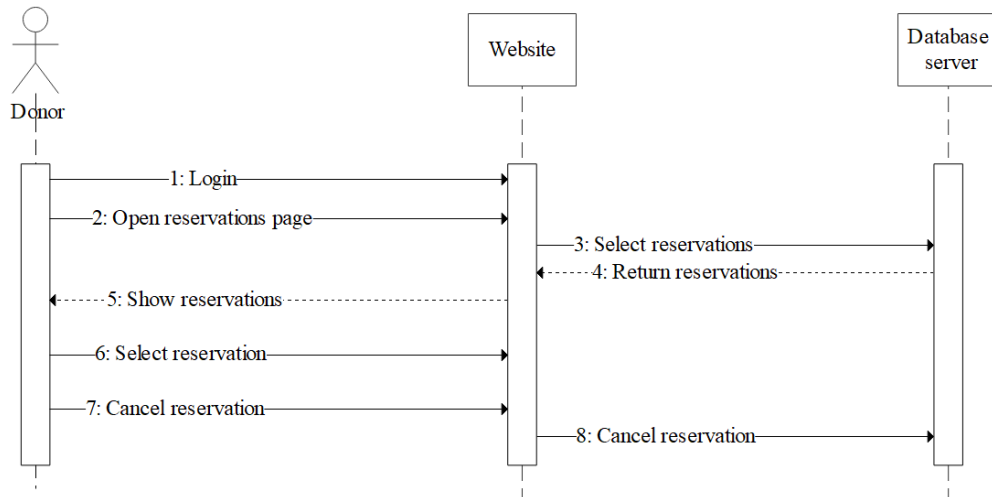


Figure 4.16 Cancel reservation sequence diagram

Figure 4.17 shows the view reservations sequence diagram, in this diagram the messages and replies between system components in view reservations are shown

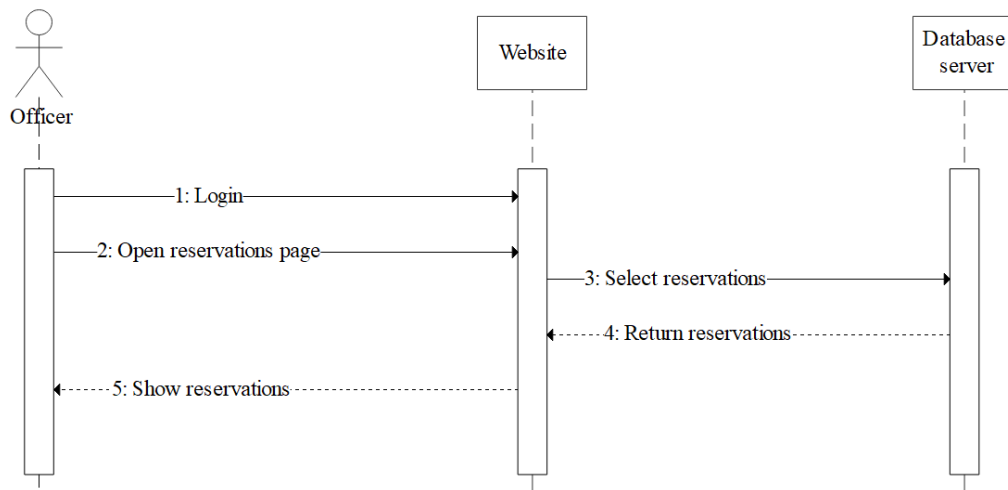


Figure 4.17 View reservations sequence diagram

4.3 Summary:

This chapter is devoted to analyzing the proposed system, in which the proposed system was initially described, the users of the system were identified, then the functional and non-functional requirements were found, in addition to defining the technical requirements for the project. Finally, the use case diagram, context diagram, and sequence diagram were created.

Chapter Five: System Design

5.1 Introduction:

The design phase of the system is one of the main phases in SDLC, in the design phase, the ER Diagram and database tables will be created, in addition to the design of prototype interfaces to the system.

5.2 ER Diagram:

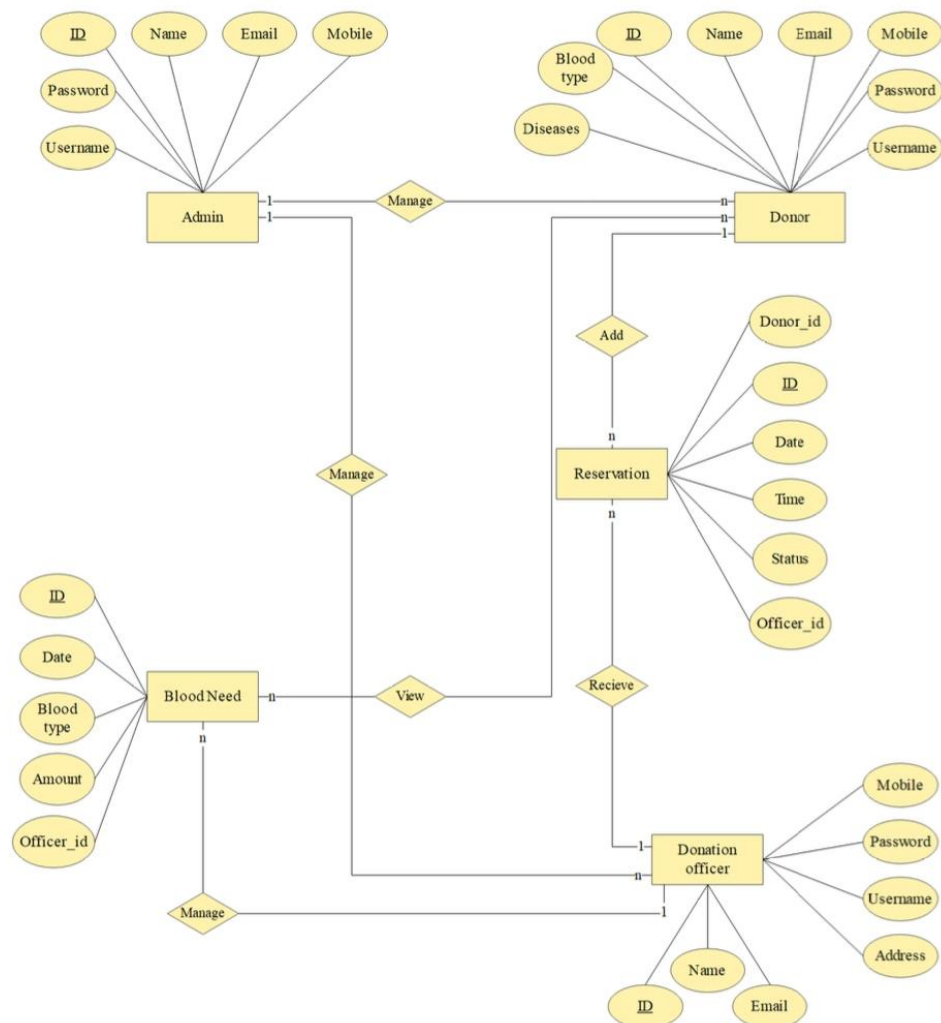


Figure 5.1 ER Diagram

5.3 Database Schema:

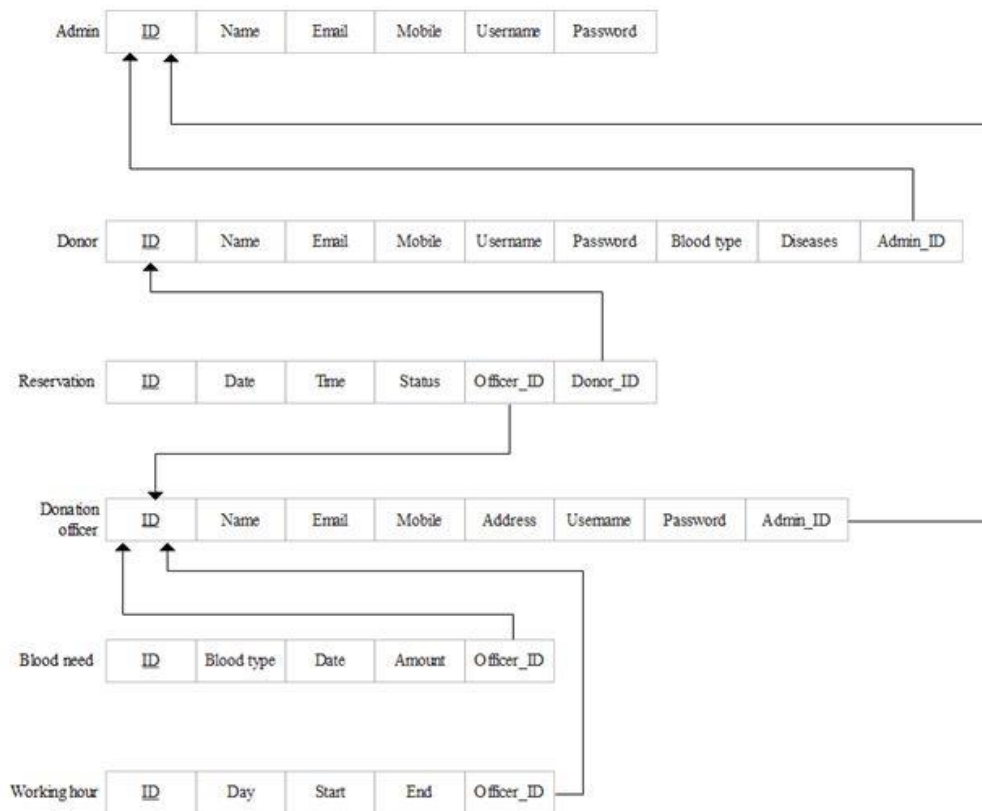


Figure 5.2 Database Schema

5.4 Database Tables:

Table 5.1 Admin table

Field name	Data type	key
ID	Int	Primary
Name	Text	
Mobile	Int	
Email	Text	
Username	Text	
Password	Text	

Table 5.2 Donor table

Field name	Data type	Key
ID	Int	Primary
Name	Text	
Mobile	Int	
Email	Text	
Blood type	Text	
Diseases	Text	
Username	Text	
Password	Text	
Admin_ID	Int	Foreign

Table 5.3 Officer table

Field name	Data type	Key
ID	Int	Primary
Name	Text	
Mobile	Int	
Email	Text	
Address	Text	
Username	Text	
Password	Text	
Admin_ID	Int	Foreign

Table 5.4 Need table

Field name	Data type	Key
ID	Int	Primary
Officer_ID	Int	Foreign
Date	Date	
Blood type	Text	

Amount	Int	
--------	-----	--

Table 5.5 Reservation table

Field name	Data type	Key
ID	Int	Primary
Officer_ID	Int	Foreign
Donor_ID	Int	Foreign
Date	Date	
Time	Time	
Status	Text	

Table 5.6 Working hour table

Field name	Data type	Key
ID	Int	Primary
Officer_ID	Int	Foreign
Day	Test	
Start	Time	
End	Time	

5.5 System Interfaces:

In this section we will design the prototype interfaces for the proposed system.

5.5.1 Interfaces for all users

The following interfaces are used by all users.

Figure 5.3 shows the main interface of the system, it shows the services that the system provides them, and information about the website.

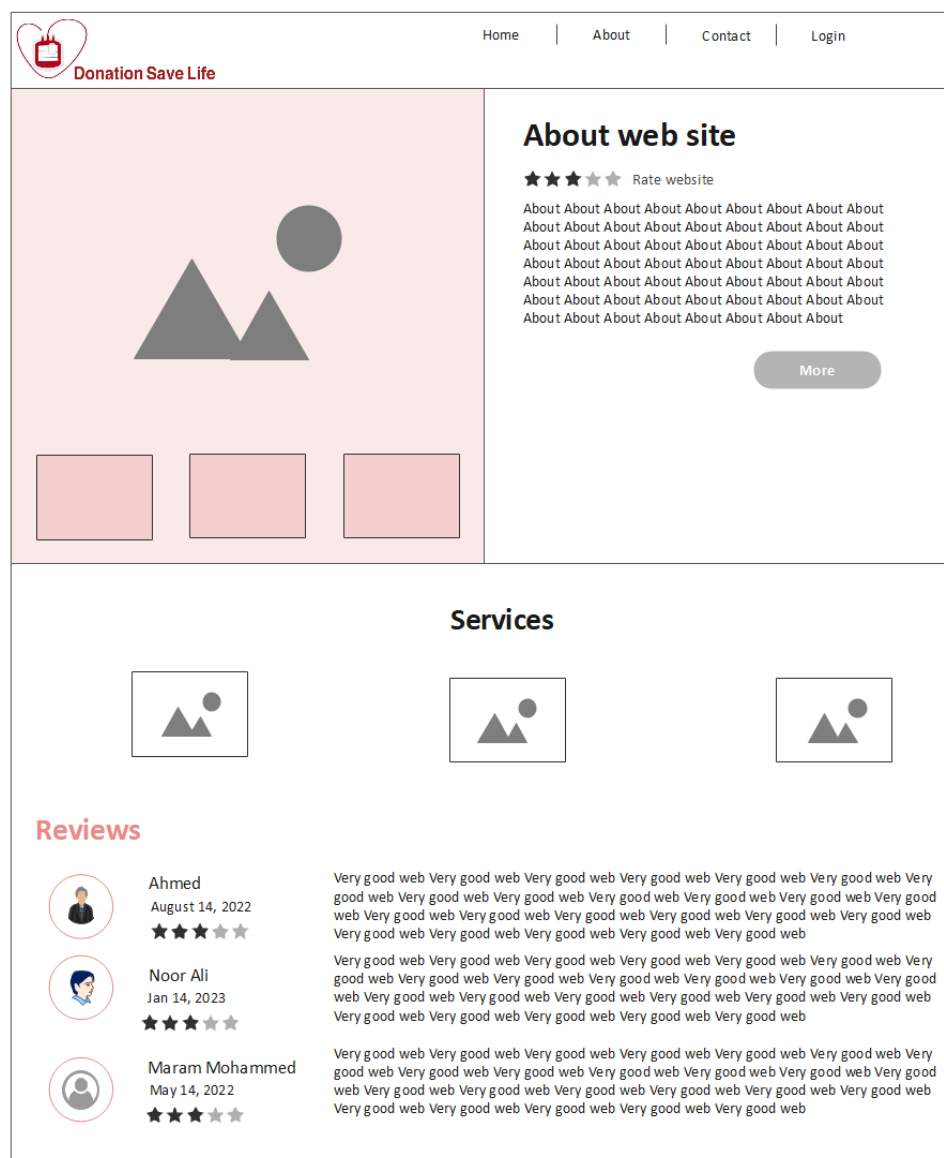
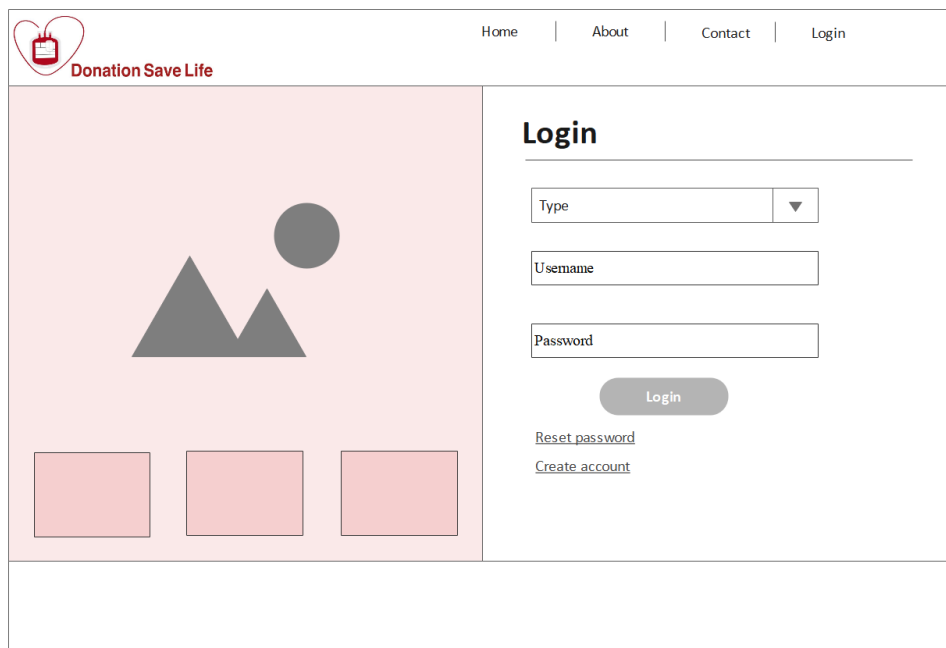


Figure 5.3 Home screen

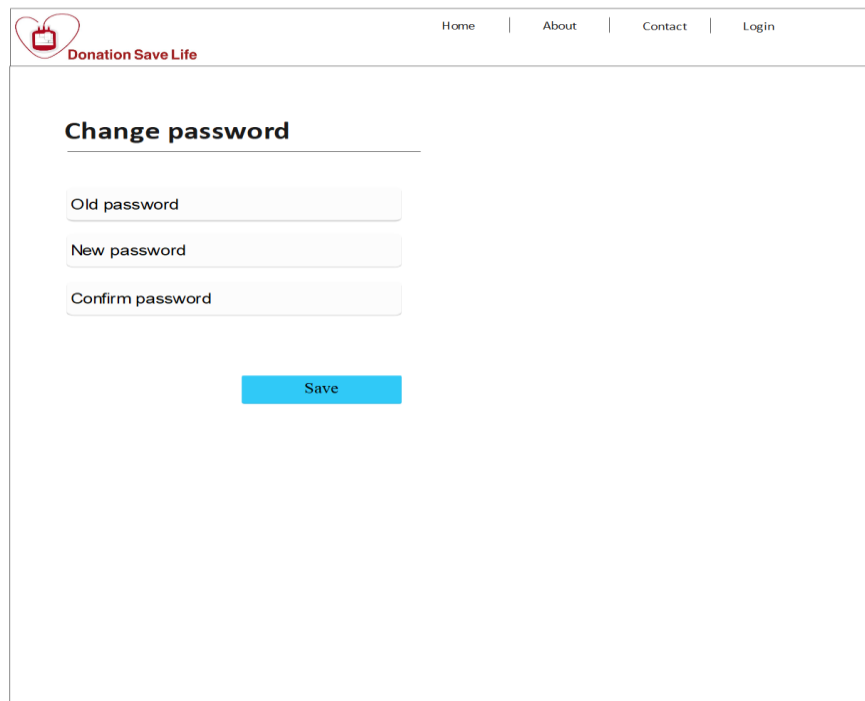
Figure 5.4 shows the login screen, where the users can login to their account using username and password.



The screenshot shows the login interface for 'Donation Save Life'. The header includes a logo with a heart and a first aid kit, and navigation links for Home, About, Contact, and Login. The main content area is split: the left side has a light pink background with a stylized mountain and sun icon and three red rectangular boxes; the right side is white and contains the 'Login' form. The form includes a 'Type' dropdown menu, 'Username' and 'Password' text input fields, a 'Login' button, and links for 'Reset password' and 'Create account'.

Figure 5.4 Login screen

Figure 5.5 shows the change password screen, this screen allows all users of the system to change their account password.



The screenshot shows the 'Change password' interface for 'Donation Save Life'. The header is identical to the login screen. The main content area is white and features the 'Change password' title. Below the title are three text input fields labeled 'Old password', 'New password', and 'Confirm password'. A blue 'Save' button is positioned at the bottom of the form.

Figure 5.5 Change password screen

5.5.2 Donor Interfaces

The following interfaces are used by donor.

Figure 5.6 shows the main screen of donor.

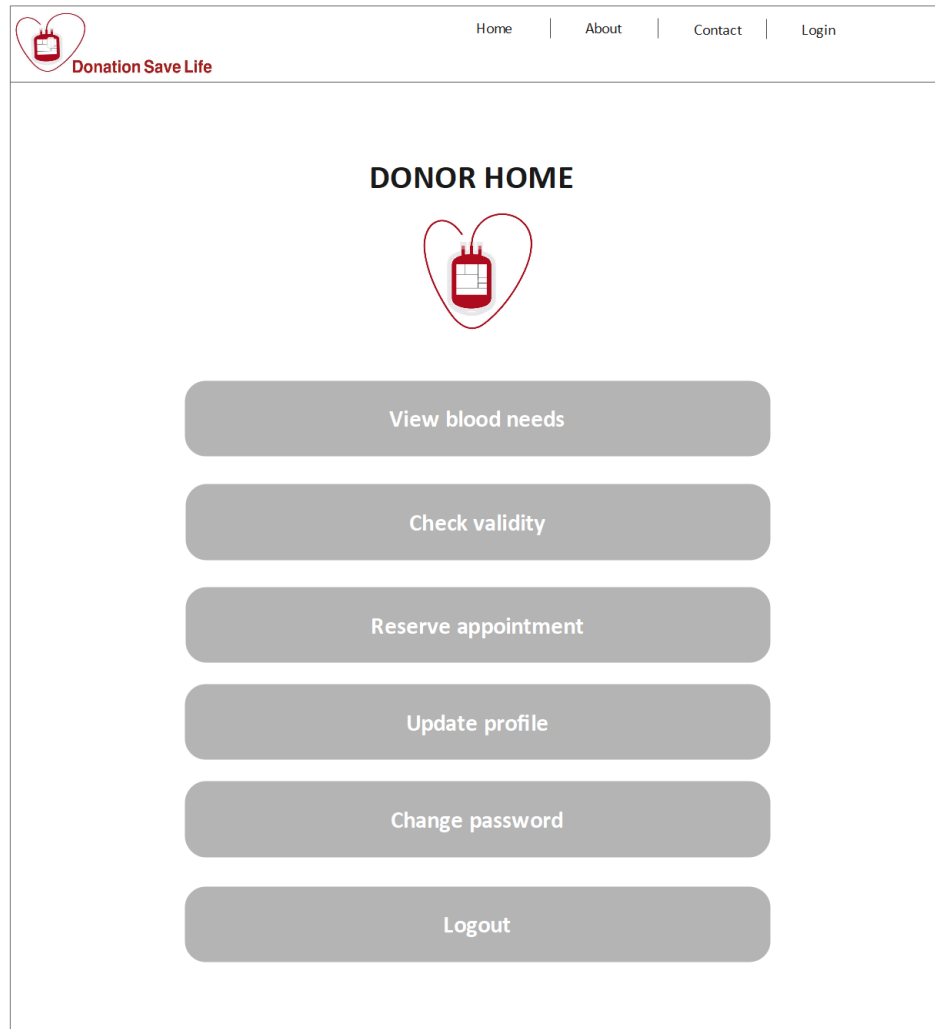


Figure 5.6 Donor home screen

Figure 5.7 shows the register screen, where the donor can create new account in the system.

The screenshot displays a web application interface for donor registration. The header features a logo with a heart and a blood drop icon, labeled 'Donation Save Life', and a navigation menu with links for Home, About, Contact, and Login. The main section is titled 'Register' and contains a form with the following fields: Name, Mobile, Email, Blood Type (a dropdown menu), Diseases, Username, and Password. A prominent blue button labeled 'Register' is positioned below the form fields.

Figure 5.7 Donor Register screen

Figure 5.8 shows the blood need screen, the donors can view all blood needs that were added by the officers.

 [Home](#) | [About](#) | [Contact](#) | [Login](#)

Blood needs

Date: 10/10/2022
Type: A+
Amount: 20 units

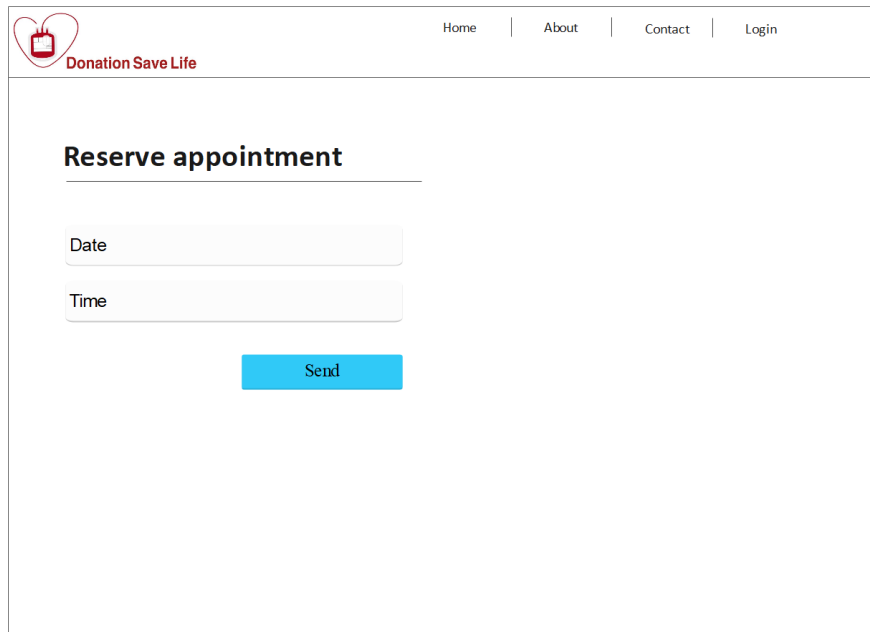
Date: 20/10/2022
Type: B+
Amount: 20 units

Date: 20/10/2022
Type: AB+
Amount: 30 units

Date: 29/10/2022
Type: O+
Amount: 5 units

Figure 5.8 Blood need screen

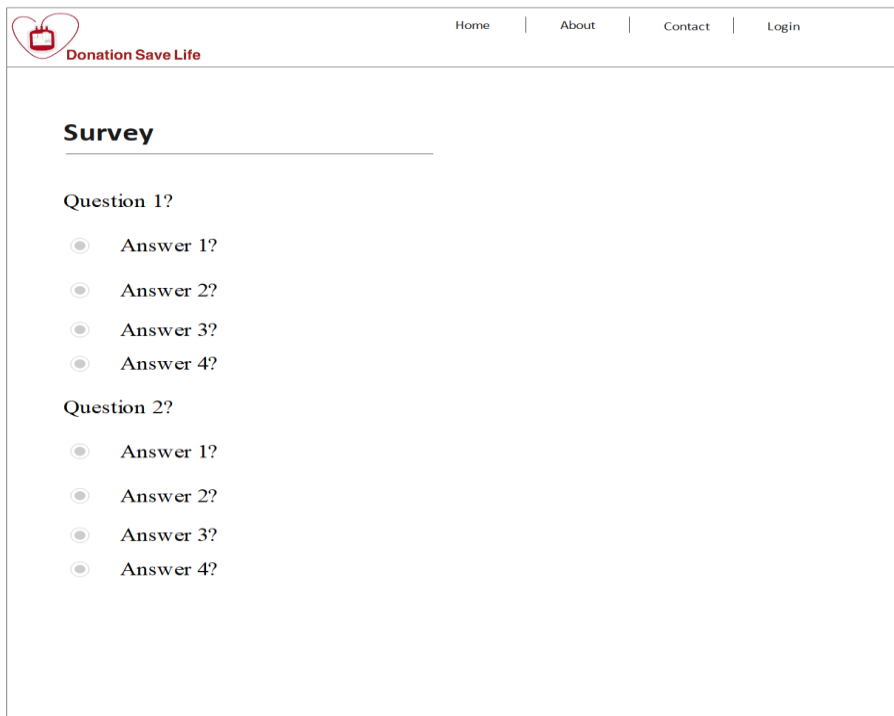
Figure 5.9 shows the reserve appointment screen, this screen allows donor to reserve appointment for blood donation.



The screenshot shows a web interface for 'Donation Save Life'. At the top, there is a navigation bar with links for Home, About, Contact, and Login. The main heading is 'Reserve appointment'. Below this, there are two input fields labeled 'Date' and 'Time'. A blue 'Send' button is positioned below the input fields.

Figure 5.9 Reserve appointment screen

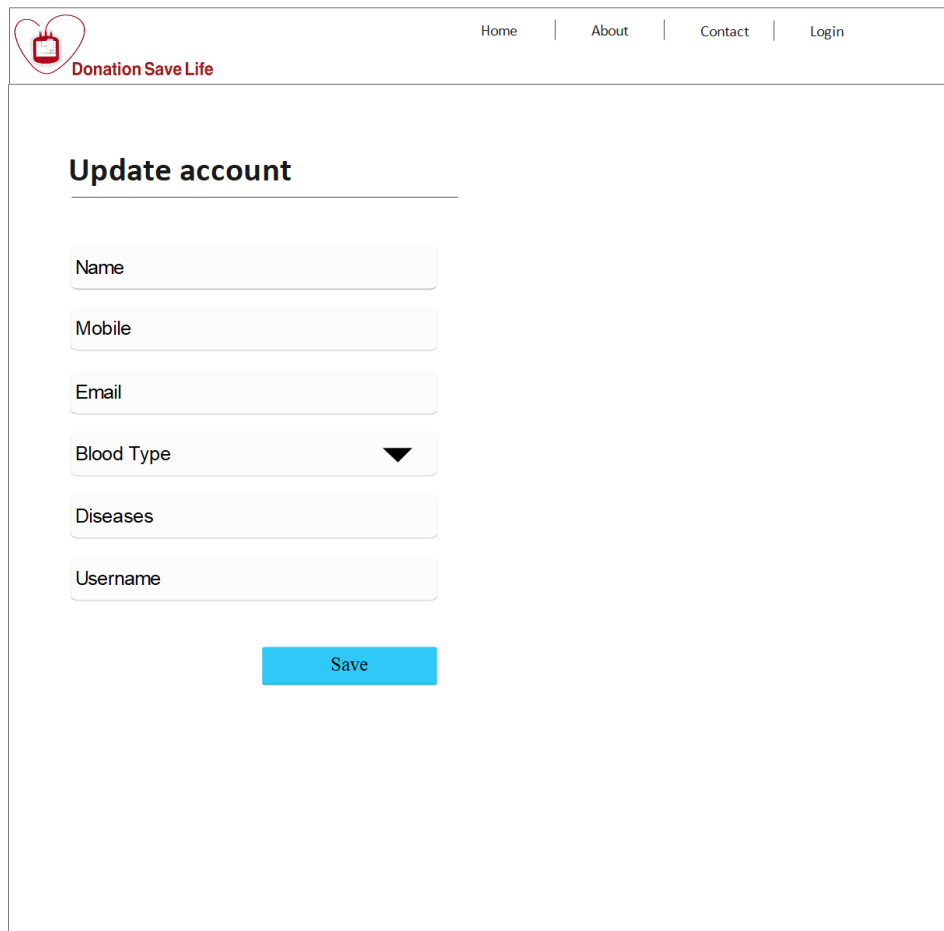
Figure 5.10 shows the survey screen, in this screen the donor can check the validity for blood donation.



The screenshot shows a web interface for 'Donation Save Life'. At the top, there is a navigation bar with links for Home, About, Contact, and Login. The main heading is 'Survey'. Below this, there are two sections: 'Question 1?' and 'Question 2?'. Each section contains four radio button options labeled 'Answer 1?', 'Answer 2?', 'Answer 3?', and 'Answer 4?'.

Figure 5.10 Survey screen

Figure 5.11 shows the update donor account screen, this screen allows all donors to update their account information.



The screenshot displays a web interface for 'Donation Save Life'. At the top, there is a navigation bar with a logo on the left and links for 'Home', 'About', 'Contact', and 'Login' on the right. The main content area is titled 'Update account' and contains a form with the following fields: 'Name', 'Mobile', 'Email', 'Blood Type' (a dropdown menu), 'Diseases', and 'Username'. A blue 'Save' button is positioned below the form fields.

Figure 5.11 Update donor account screen

5.5.3 Officer interfaces

The following interfaces are used by officer.

Figure 5.12 shows the main screen of officer.

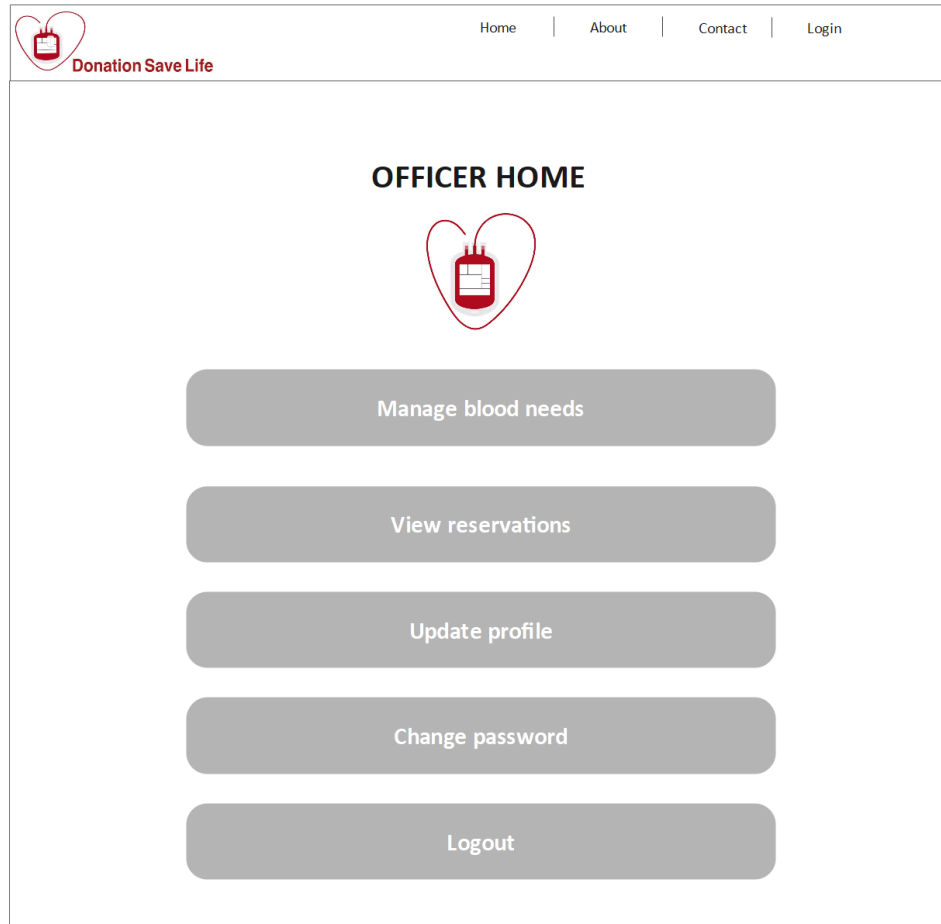


Figure 5.12 Officer main screen

Figure 5.13 shows reservation screen, this screen allows officer to view all donation appointments that reserved by donors.

The screenshot shows the 'Reservations' screen of the 'Donation Save Life' application. The header includes a logo with a heart and a calendar icon, the text 'Donation Save Life', and navigation links for Home, About, Contact, and Login. The main content area is titled 'Reservations' and displays a list of four reservation entries, each in a separate box:

- Date: 10/10/2022
Time: 10:30
Donor: Noor
- Date: 20/10/2022
Type: B+
Amount: 20 units
- Date: 10/10/2022
Time: 11:30
Donor: Maram
- Date: 20/10/2022
Time: 12:30
Donor: Hessah

Figure 5.13 Reservations screen

Figure 5.14 shows the added blood need screen, where the officer can add new blood need.

The screenshot shows the 'Add need' screen of the 'Donation Save Life' application. The header is identical to the previous screen. The main content area is titled 'Add need' and contains three input fields: 'Date', 'Amount', and 'Blood Type' (which is a dropdown menu). Below these fields is a blue 'Add' button.

Figure 5.14 Add need screen

Figure 5.15 shows update officer profile screen, this screen allows officer to update the information of account.

The screenshot displays the 'Update account' interface. At the top, the 'Donation Save Life' logo is on the left, and navigation links for 'Home', 'About', 'Contact', and 'Login' are on the right. The main content area is titled 'Update account' and contains five text input fields labeled 'Name', 'Mobile', 'Email', 'Address', and 'Username'. A blue 'Save' button is positioned below the input fields.

Figure 5.15 Update officer account screen

5.5.4 Admin Interfaces

The following interfaces are used by admin.

Figure 5.16 shows the first admin screen.

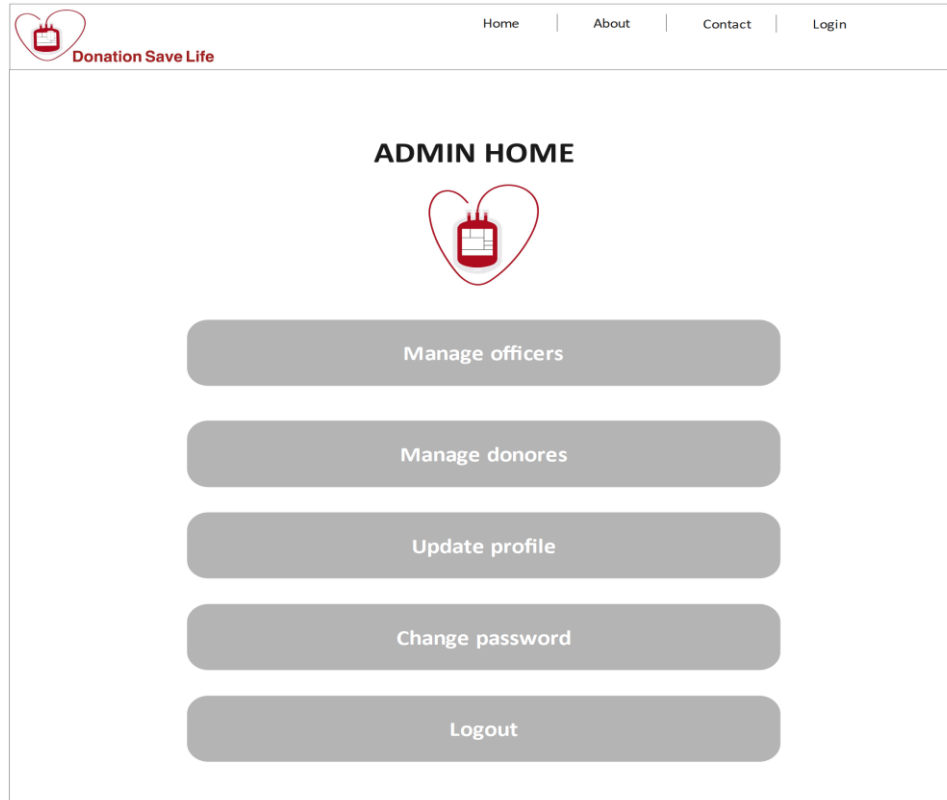


Figure 5.16 Admin home screen

Figure 5.17 shows manage donors' screen this screen allows admin of the system to manage all donors accounts.

 Donation Save Life

Home | About | Contact | Login

Manage donors

Name	Email	Mobile	Type	Delete
Nada Ali	Nnn@gmail.com	0554123365	Donor	Delete
Khalid ahmed	k123@gmail.com	0541236548	Donor	Delete
Noor Ali	noor@gmail.com	0541366939	Donor	Delete
Abdullah Saad	a123@hotmail.com	0545147899	Donor	Delete
Reem Mohammed	r123@gmail.com	0596352418	Donor	Delete
Dalal Saa'd	dalal@hotmail.com	0501239875	Donor	Delete

Figure 5.17 Manage donors' screen

Figure 5.18 shows manage officers' screen. This screen allows admin of the system to manage all officers' accounts.



Home | About | Contact | Login

Manage officers

Name	Email	Mobile	Type	<u>Delete</u>
Ahmed Ali	ah123@gmail.com	0554127845	Officer	Delete
Noor Abdullah	nana2223@gmail.com	0541231212	Officer	Delete
Hessah Khalid	hk1234@gmail.com	0541367415	Officer	Delete

Figure 5.18 Manage officers' screen

Figure 5.19 shows update admin account screen, this screen allows all admin of the system to manage all officers accounts.

The screenshot displays a web interface for updating an admin account. At the top, a header bar contains a logo with a heart and a first aid kit, labeled 'Donation Save Life', and a navigation menu with links for 'Home', 'About', 'Contact', and 'Login'. The main section is titled 'Update account' and contains four text input fields for 'Name', 'Mobile', 'Email', and 'Username'. A blue 'Save' button is located at the bottom right of the form area.

Figure 5.19 Update admin account screen

5.6 Summary:

In this chapter we have completed the design phase of the proposed system, in this phase we start from ER diagram design, the database tables creation, and finally we designed prototype interfaces for the proposed system.

Chapter Six:

System Implementation

6.1 Introduction

In this chapter the system will be implemented, the database of the proposed system will be created using MySQL server, and system interfaces will be created using PHP, HTML programming language.

6.2 Database Implementation

The next tables are the database tables.

Figure 6.1 Admin table This this database table for storing admin account information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop More
☐	2	name	varchar(50)	utf8_bin		No	None			 Change  Drop More
☐	3	email 	varchar(60)	utf8_bin		No	None			 Change  Drop More
☐	4	mobile 	varchar(10)	utf8_bin		No	None			 Change  Drop More
☐	5	username 	varchar(15)	utf8_bin		No	None			 Change  Drop More
☐	6	password	varchar(15)	utf8_bin		No	None			 Change  Drop More

Figure 6.1 Admin table

Figure 6.2 Donor table This this database table for storing donor account information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop More
☐	2	name	varchar(50)	utf8_bin		No	None			 Change  Drop More
☐	3	email 	varchar(60)	utf8_bin		No	None			 Change  Drop More
☐	4	mobile 	varchar(10)	utf8_bin		No	None			 Change  Drop More
☐	5	birthdate	date			No	None			 Change  Drop More
☐	6	diseases	varchar(200)	utf8_bin		No	None			 Change  Drop More
☐	7	blood_type	varchar(3)	utf8_bin		No	None			 Change  Drop More
☐	8	username 	varchar(15)	utf8_bin		No	None			 Change  Drop More
☐	9	password	varchar(15)	utf8_bin		No	None			 Change  Drop More

Figure 6.2 Donor table

Figure 6.3 Officer table This this database table for storing officer account information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2	name	varchar(50)	utf8_bin		No	None			Change Drop More
☐	3	email	varchar(60)	utf8_bin		No	None			Change Drop More
☐	4	mobile	varchar(10)	utf8_bin		No	None			Change Drop More
☐	5	address	varchar(200)	utf8_bin		No	None			Change Drop More
☐	6	username	varchar(15)	utf8_bin		No	None			Change Drop More
☐	7	password	varchar(15)	utf8_bin		No	None			Change Drop More

Figure 6.3 Officer table

Figure 6.4 Working hour table This this database table for storing working hour of officer information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2	day	varchar(10)	utf8_bin		No	None			Change Drop More
☐	3	start	time			No	None			Change Drop More
☐	4	end	time			No	None			Change Drop More
☐	5	officer_id	int(11)			No	None			Change Drop More

Figure 6.4 Working hour table

Figure 6.5 Needs table This this database table for storing blood needs information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop More
☐	2	amount	int(11)			No	None			 Change  Drop More
☐	3	bood_type	varchar(4)	utf8_bin		No	None			 Change  Drop More
☐	4	datetime	datetime			No	None			 Change  Drop More
☐	5	officer_id	int(11)			No	None			 Change  Drop More

Figure 6.5 Needs table

Figure 6.6 Reservation table This this database table for storing reservations information.

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id 	int(11)			No	None		AUTO_INCREMENT	 Change  Drop More
☐	2	date	date			No	None			 Change  Drop More
☐	3	time	time			No	None			 Change  Drop More
☐	4	officer_id	int(11)			No	None			 Change  Drop More
☐	5	donor_id	int(11)			No	None			 Change  Drop More
☐	6	status	varchar(15)	utf8_bin		No	None			 Change  Drop More

Figure 6.6 Reservation table

6.3 System Interfaces

The system interfaces were created using PHP, HTML, CSS proگرامing languages.

6.3.1 Public Interfaces

Figure 6.7 Home screen This is the screen security of the system.

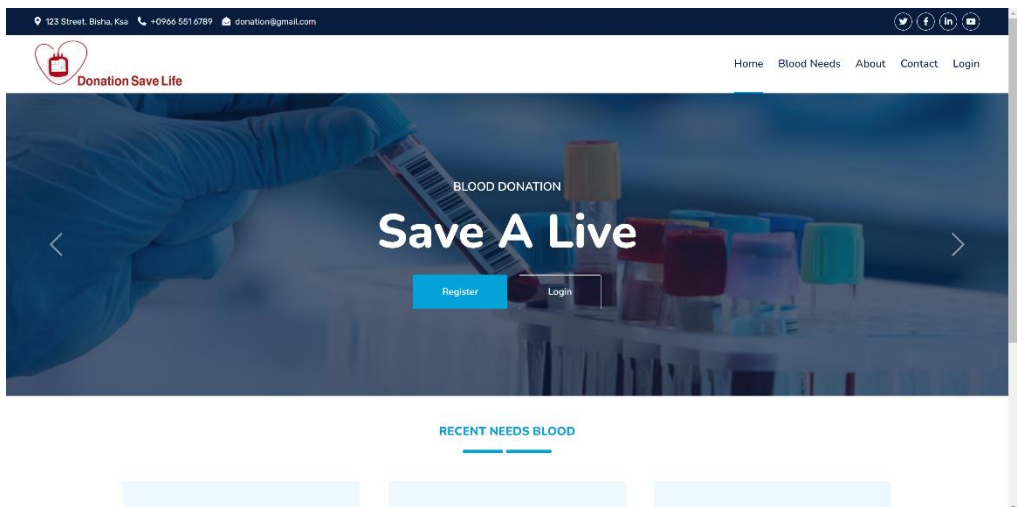


Figure 6.7 Home screen

Figure 6.8 All needs screen.This screen shoes the blood needs that have been added from the different officers in this system.

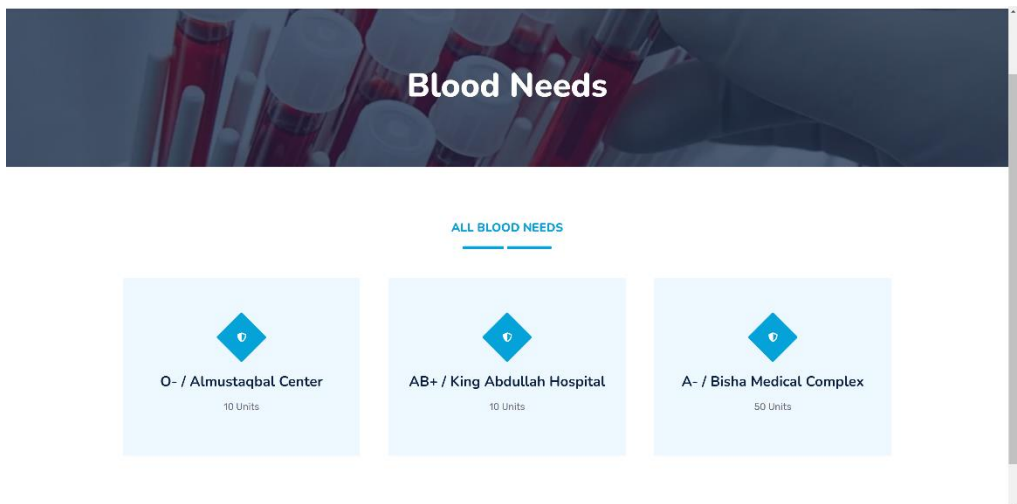


Figure 6.8 All needs screen.

Figure 6.9 About screen This screen shows a brief description about the system.

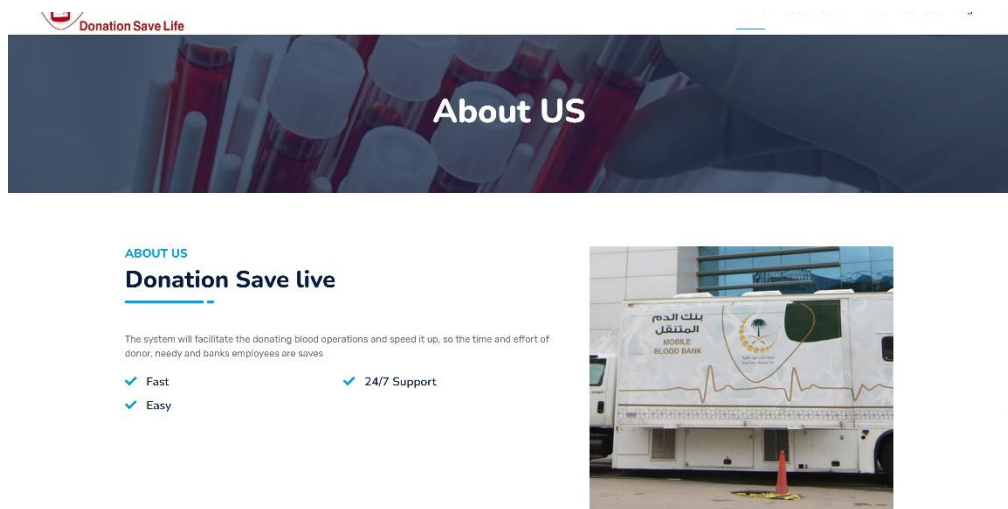


Figure 6.9 About screen

Figure 6.10 Contact us screen This screen shows contact information of the system, phone, e-mail and address.

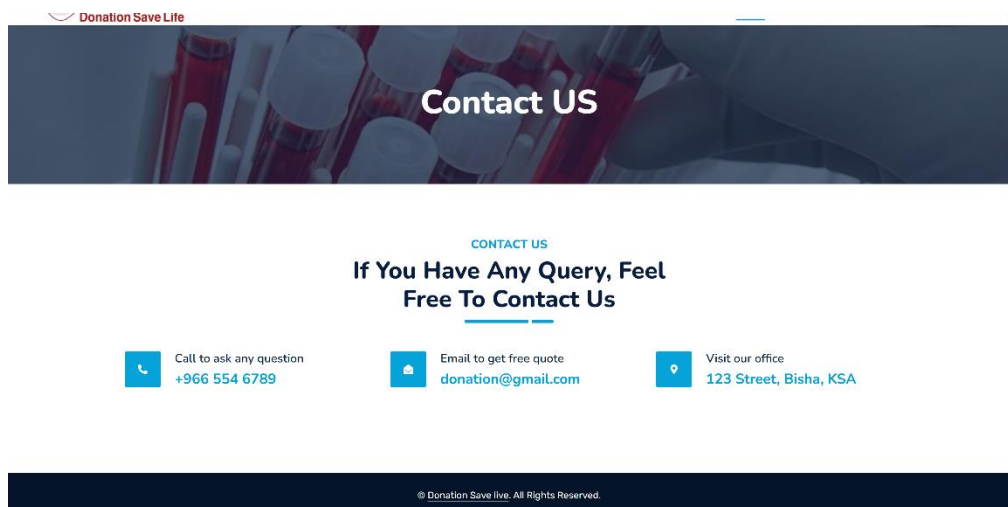


Figure 6.10 Contact us screen

Figure 6.11 Login screen This screen allows different users to log into their accounts using username and password.

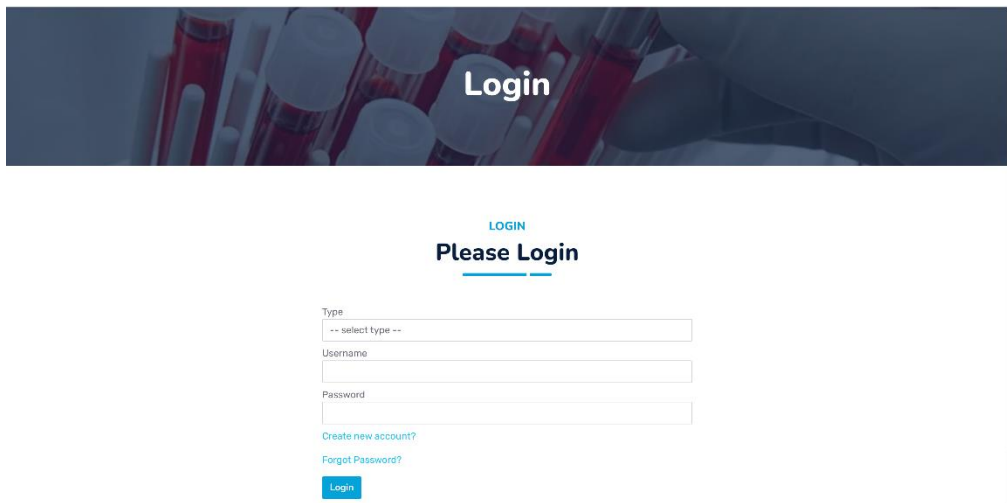
The screenshot shows a web application's login interface. At the top, there is a dark blue header with the word "Login" in white. Below this, the text "LOGIN" is in small blue letters, followed by "Please Login" in bold black. The form includes a "Type" dropdown menu with "-- select type --", a "Username" text input, and a "Password" text input. Below the password field are two links: "Create new account?" and "Forgot Password?". A blue "Login" button is at the bottom of the form. The background of the page features a blurred image of medical test tubes.

Figure 6.11 Login screen

Figure 6.12 Forget passwords screen This screen allows users to reset their passwords in case of the forget them.

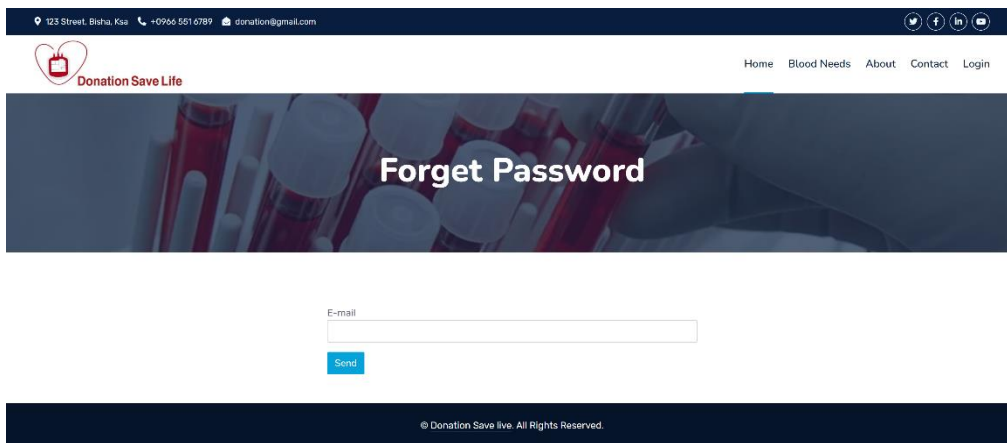
The screenshot displays the "Forget Password" screen. At the top, a dark blue navigation bar contains contact information (address, phone, email) and social media icons. Below this is a white header with the "Donation Save Life" logo and a navigation menu (Home, Blood Needs, About, Contact, Login). The main content area has a dark blue header with "Forget Password" in white. Below this is a form with an "E-mail" text input and a blue "Send" button. The footer is a dark blue bar with the copyright notice "© Donation Save live. All Rights Reserved.".

Figure 6.12 Forget passwords screen

Figure 6.13 Account type screen This screen allow user to select the type of account she wants to create.

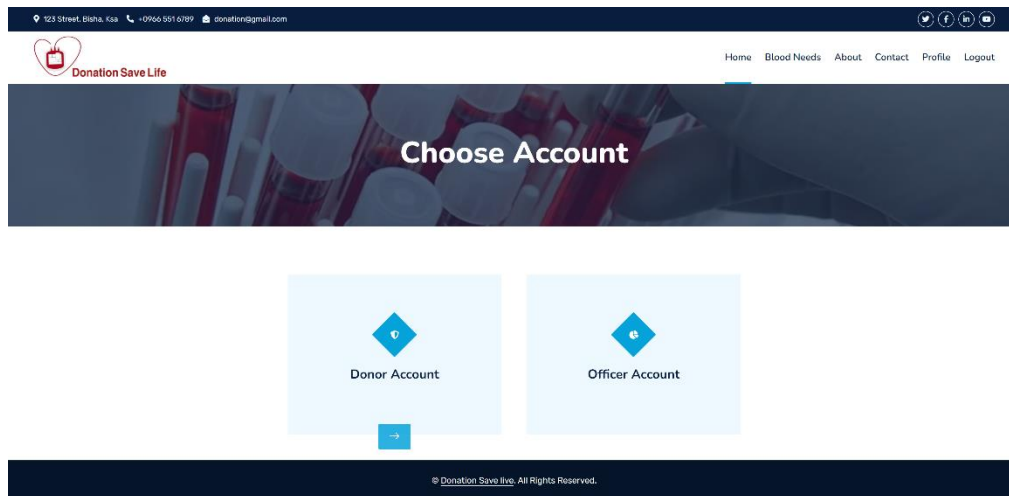


Figure 6.13 Account type screen

6.3.2 Admin Interfaces

Figure 6.14 Manage donors screen This screen allows the System Administrator to view all donors that have been registered in the system, she can delete.

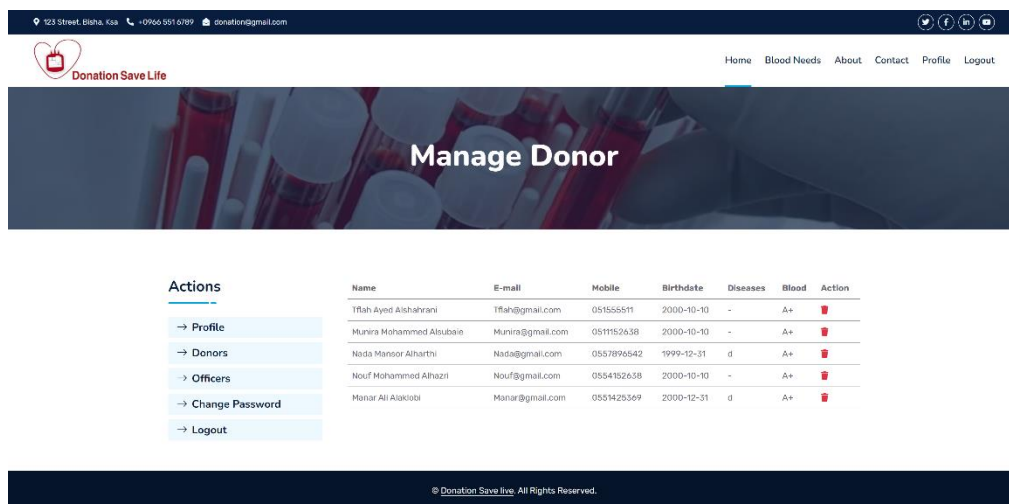


Figure 6.14 Manage donors screen

Figure 6.15 Manage officers' screen This screen allows the System Administrator to view all officers that have been registered in the system, she can delete.

123 Street, Bisha, KSA | 0966 9516789 | donation@gmail.com

Home | Blood Needs | About | Contact | Profile | Logout

Manage Officer

Actions

- Profile
- Donors
- **Officers**
- Change Password
- Logout

Name	E-mail	Mobile	Address	Action
Almustaqbal Center	Almustaqbal@gmail.com	0557895286	255, Al Nakhil, Bisha 67716, Saudi Arabia	
Bisha Medical Complex	Complex@gmail.com	0554162639	2365+28, ????????? 7771 6771, Saudi Arabia	
King Abdullah Hospital	King@gmail.com	0500000002	Al Nakhil 6956 4778 7771 ????? 77, Bisha 67714, Saudi Arabia	

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Figure 6.15 Manage officers' screen

Figure 6.16 Update admin account screen This screen allows the System Administrator to update here account information.

123 Street, Bisha, KSA | 0966 9516789 | donation@gmail.com

Home | Blood Needs | About | Contact | Profile | Logout

Profile

Actions

- **Profile**
- Donors
- Officers
- Change Password
- Logout

Name
admin name

E-mail
admin@gmail.com

Mobile
0554162879

Username
admin

Update

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Figure 6.16 Update admin account screen

Figure 6.17 Change admin password screen This screen allows administrator of the system to change her account password.

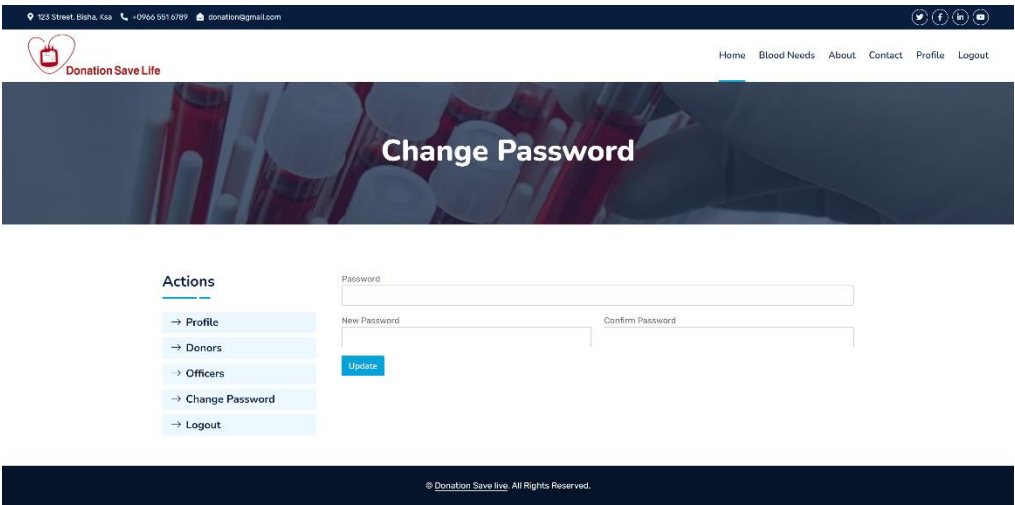


Figure 6.17 Change admin password screen

6.3.3 Officer Interfaces

Figure 6.18 Manage blood needs screen This screen allow the officer to view all blood needs that she have been added to the system, she can delete any of them.

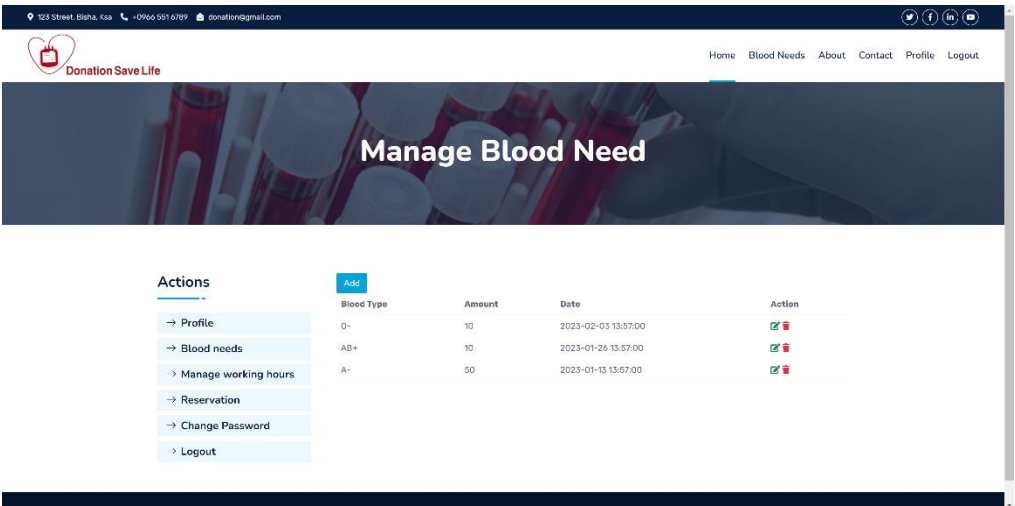


Figure 6.18 Manage blood needs screen

Figure 6.19 Add need screen This screen allow officer to add and you blood need to the system.

The screenshot shows the 'Add Blood Need' screen of the 'Donation Save Life' application. The header includes the application name, contact information, and navigation links. The main content area features a sidebar with 'Actions' (Profile, Blood needs, Manage working hours, Reservation, Change Password, Logout) and a form for adding a new blood need. The form fields are: Blood Type (dropdown menu), Amount (text input), and Date (calendar icon). A blue 'Add' button is located below the form.

123 Street, Bisha, KSA | +0966 551 61789 | donation@gmail.com

Home | Blood Needs | About | Contact | Profile | Logout

Add Blood Need

Actions

- Profile
- Blood needs
- Manage working hours
- Reservation
- Change Password
- Logout

Blood Type
-- select blood type --

Amount

Date
mm/dd/yyyy -->

Add

Figure 6.19 Add need screen

Figure 6.20 Update need screen This screen allow officer to update in formation of a blood need.

The screenshot shows the 'Update Blood Need' screen of the 'Donation Save Life' application. The header and sidebar are identical to Figure 6.19. The form for updating a blood need is displayed, with fields for Blood Type (dropdown menu), Amount (text input), and Date (calendar icon). A blue 'Edit' button is located below the form.

123 Street, Bisha, KSA | +0966 551 61789 | donation@gmail.com

Home | Blood Needs | About | Contact | Profile | Logout

Add Blood Need

Actions

- Profile
- Blood needs
- Manage working hours
- Reservation
- Change Password
- Logout

Blood Type
0-

Amount
10

Date
02/03/2023 13:57

Edit

Figure 6.20 Update need screen

Figure 6.21 Manage working hours screen This screen allow officer to view all walking hours, she can add delete or update any of them.

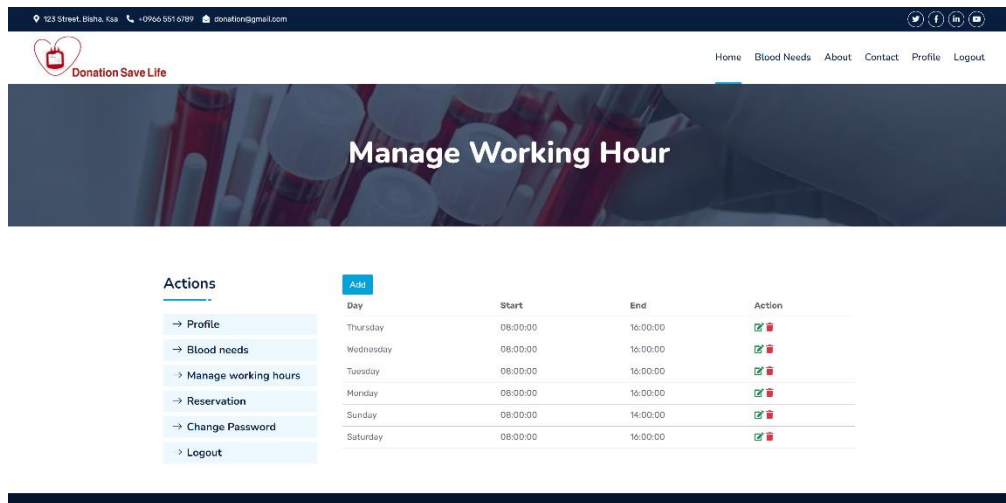


Figure 6.21 Manage working hours screen

Figure 6.22 Add working hour screen This screen allow officer to add new working hour.

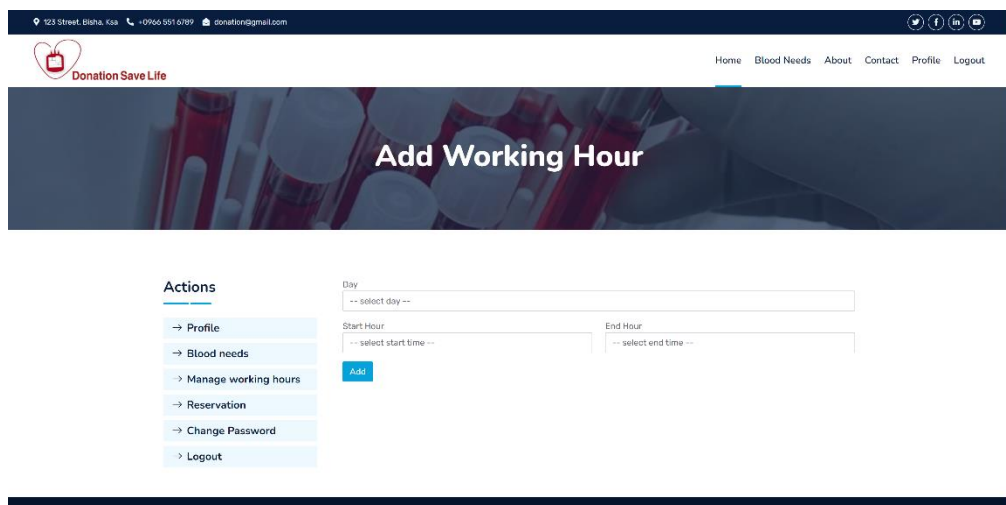


Figure 6.22 Add working hour screen

Figure 6.23 Update working hour screen This screen allow officer to update a walking hour information.

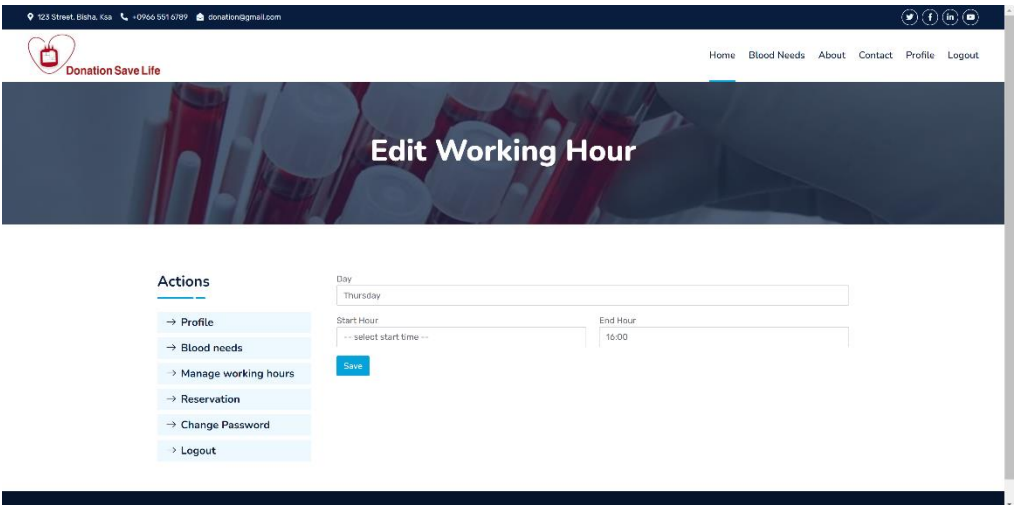


Figure 6.23 Update working hour screen

Figure 6.24 Manage reservations screen This screen allow officer to view all reservations that have been added from donors, she can accept or reject any of them.

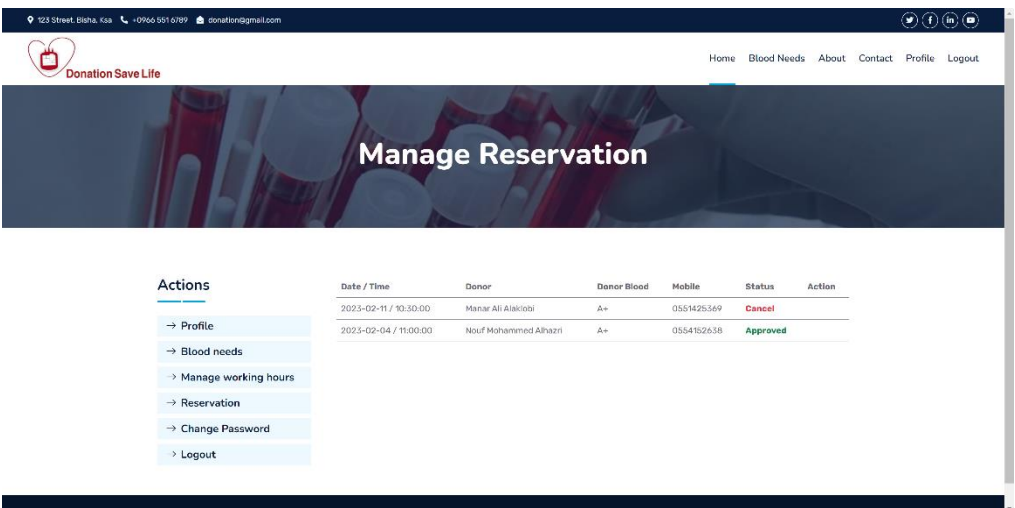


Figure 6.24 Manage reservations screen

Figure 6.25 Update officer account screen This screen allow officer to update her account information.

The screenshot shows the 'Profile' update interface. At the top, there is a dark blue header with contact information: '123 Street, Bisha, KSA', '+0966 951 01799', and 'donation@gmail.com'. Below this is a navigation bar with links: Home, Blood Needs, About, Contact, Profile, and Logout. The main header area features the 'Donation Save Life' logo and a large banner with the word 'Profile' in white text. The content area is divided into two sections. On the left, an 'Actions' sidebar lists: Profile (selected), Blood needs, Manage working hours, Reservation, Change Password, and Logout. On the right, a form contains the following fields: Name (King Abdullah Hospital), E-mail (King@gmail.com), Mobile (0500000002), Address (Al Nakhjil 4950-4778, Bisha 67714, Saudi Arabia), and Username (King123). A blue 'Update' button is positioned at the bottom right of the form.

Figure 6.25 Update officer account screen

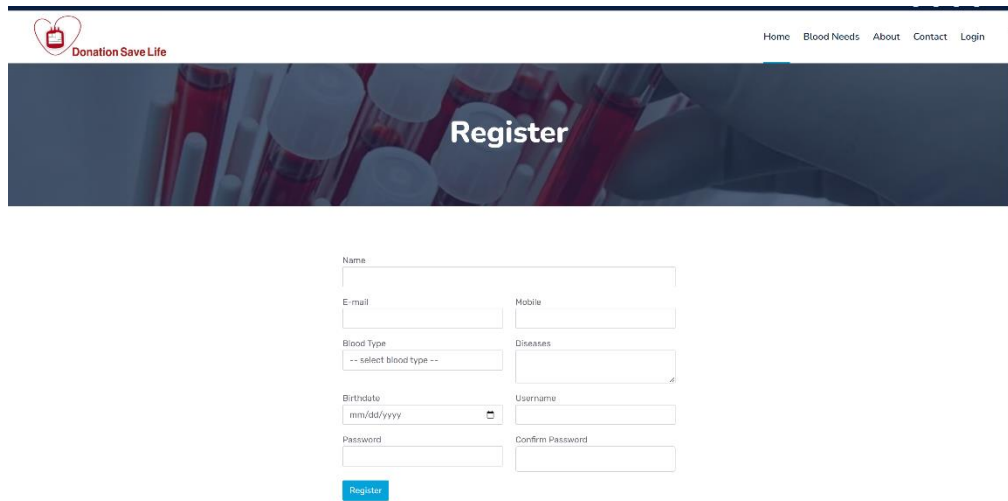
Figure 6.26 Change officer password screen This screen allow officer to update her account password.

The screenshot shows the 'Change Password' interface. It features the same dark blue header and navigation bar as Figure 6.25. The main banner area displays the text 'Change Password' in white. The 'Actions' sidebar on the left remains the same, with 'Change Password' now selected. The form on the right includes a 'Password' field, a 'New Password' field, and a 'Confirm Password' field. A blue 'Update' button is located below the 'New Password' field.

Figure 6.26 Change officer password screen

6.3.4 Donor Interfaces

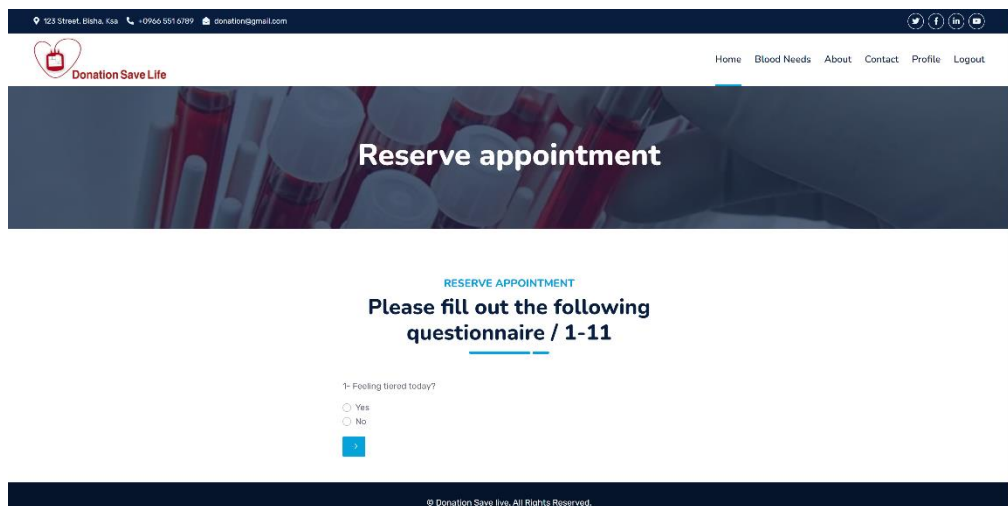
Figure 6.27 Register screen This screen allowed donor to create a new account in this system.



The screenshot shows the 'Register' page of the 'Donation Save Life' website. The header includes the logo and navigation links: Home, Blood Needs, About, Contact, and Login. The main heading is 'Register'. Below it is a registration form with the following fields: Name, E-mail, Mobile, Blood Type (a dropdown menu showing '-- select blood type --'), Diseases, Birthdate (with a date picker), Username, Password, and Confirm Password. A blue 'Register' button is located at the bottom of the form.

Figure 6.27 Register screen

Figure 6.28 Answer survey screen This screen allowed donor to answer the survey, so she can check the validity of her of her for blood donation.



The screenshot shows the 'Reserve appointment' page of the 'Donation Save Life' website. The header includes the logo and navigation links: Home, Blood Needs, About, Contact, Profile, and Logout. The main heading is 'Reserve appointment'. Below it is a section titled 'RESERVE APPOINTMENT' with the instruction 'Please fill out the following questionnaire / 1-11'. The first question is '1- Feeling tired today?' with radio button options for 'Yes' and 'No'. A blue 'Next' button is at the bottom of the question. The footer contains the copyright notice: '© Donation Save Life, All Rights Reserved.'

Figure 6.28 Answer survey screen

Figure 6.29 Select donation center screen This screen allowed to select the donation centre to reserve appointment in it.

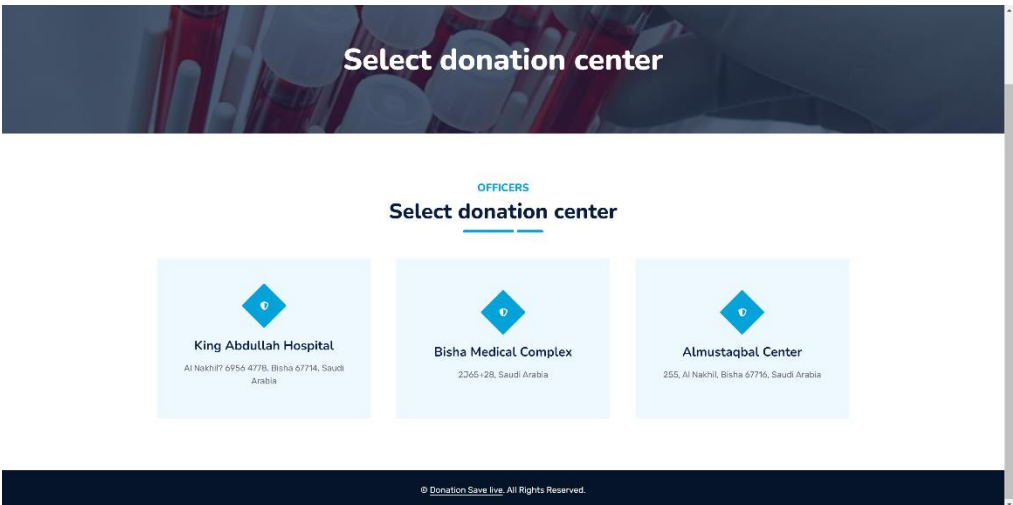


Figure 6.29 Select donation center screen

Figure 6.30 Reserve screen This screen allowed donor to reserve appointment.

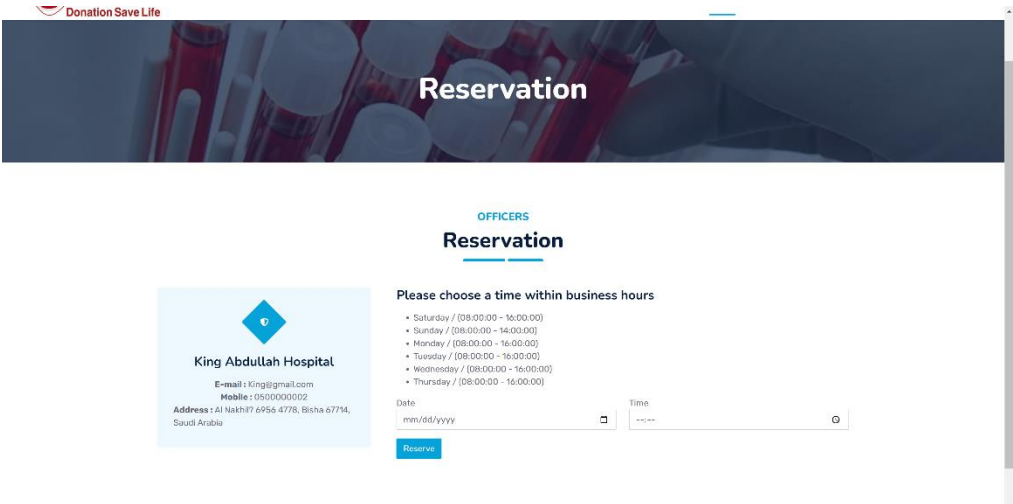


Figure 6.30 Reserve screen

Figure 6.31 Donor reservations screen This screen allowed donor to view her reservations, she can cancel any of them.

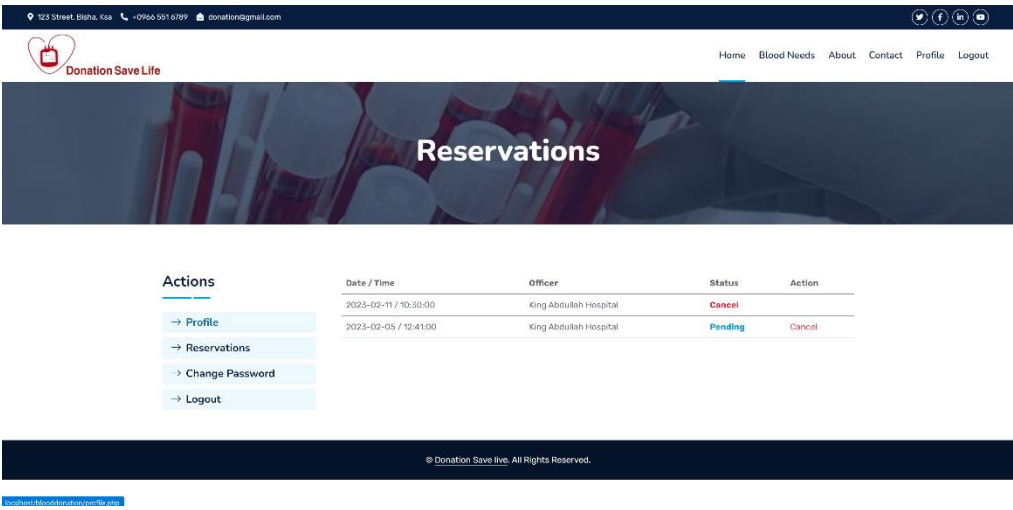


Figure 6.31 Donor reservations screen

Figure 6.32 Update donor profile screen This allowed donor to update her account in formation.

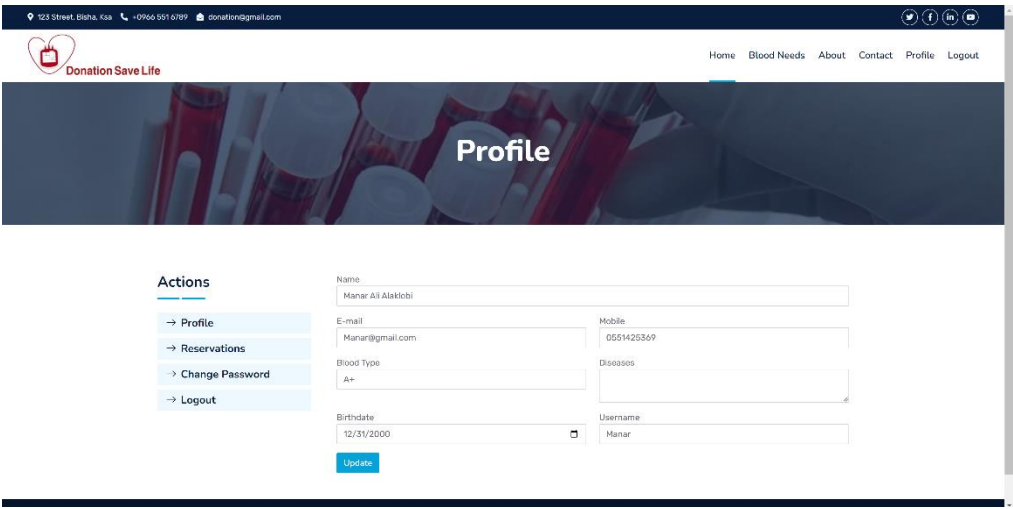


Figure 6.32 Update donor profile screen

Figure 6.33 Change donor password This screen allowed donor to update their account password.

The screenshot shows a web application interface for 'Donation Save Life'. At the top, there is a dark blue header with contact information on the left and social media icons on the right. Below the header, a navigation bar contains links: Home, Blood Needs, About, Contact, Profile, and Logout. The main content area has a large banner with the text 'Change Password' over a background image of blood test tubes. On the left side of the main area, there is a sidebar titled 'Actions' with a list of links: Profile, Reservations, Change Password (which is highlighted), and Logout. The main form area contains three input fields: 'Password', 'New Password', and 'Confirm Password'. Below these fields is a blue 'Update' button. At the bottom of the page, there is a dark blue footer with the text '© Donation Save Life. All Rights Reserved.'

Figure 6.33 Change donor password

6.4 Summary:

In this chapter we have completed the implementation phase of the proposed system.

Chapter Seven: Conclusion and Future Works

7.1 Conclusion

It is a voluntary medical procedure in which blood or one of its compounds is transferred from a healthy person to a sick person who needs blood.

Despite the growing need for blood in hospitals, blood in banks is not enough due to the small number of donors.

This project aimed to create a website to allow blood donors and blood seekers to obtain blood online.

The project helps in facilitating blood donation, developing a blood management information system to manage blood donor records, ease of blood distribution in different hospitals according to hospital requirements.

Through this project there were many challenges; managing team and time is one of these challenges, other challenges are determining the specific features of the website.

These challenges were overcome by teamwork, making timetable, do research and watch educational videos, learned lessons of this project are ability to work as team, organize time, do research and deal with new programs.

7.2 Future Works

- Create mobile application for the system.
- Support Arabic language.
- Allow donor to find nearest donation center.

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Appendix: Code

8.1 Login

```
<!DOCTYPE html>

<html lang="en">

<?php include "theme/header.php";

$title = 'Login';

if (isset($_POST['login'])) {

    $errors = array();

    $fields = array('Type'=>'type', 'Username'=>
'username', 'Password'=> 'password');

    foreach ($fields as $key=> $field) {

        if (!isset($_POST[$field]) || empty(trim($_POST[$field])))
        {

            $errors[$field] = "$key field is required";

        }

    }

    if (empty($errors)) {

        $type = mysqli_real_escape_string($conn,
trim($_POST['type']));

        $username = mysqli_real_escape_string($conn,
trim($_POST['username']));

        $password = mysqli_real_escape_string($conn,
trim($_POST['password']));

        $sql = "SELECT * FROM ` $type ` where `password`=' $password '
and `username`=' $username ' LIMIT 1";

        $result = $conn->query($sql);

        if ($result->num_rows > 0) {

            while ($row = $result->fetch_assoc()) {

                $_SESSION['user_id'] = $row['id'];

            }

        }

    }

}
```

```

        $_SESSION['type'] = $type;

        $_SESSION['table'] = $type;

        header('location:index.php');

        exit();

    }

} else {

    $errors['login'] = 'Username / Password invalid';

}

}

}

?>

<body>

<!-- Spinner Start -->

<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">

    <div class="spinner"></div>

</div>

<!-- Spinner End -->

<!-- Topbar Start -->

<?php include "theme/topbar.php"; ?>

<!-- Topbar End -->

<!-- Navbar & Carousel Start -->

<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>

```



```

        <?php include 'theme/top_page.php'; ?>

</div>

<!-- Navbar & Carousel End -->

<!-- Service Start -->

<div class="container-fluid pb-5 wow fadeInUp" data-wow-
delay="0.1s">

    <div class="container">

        <div class="section-title text-center position-relative pb-
3 mb-5 mx-auto" style="max-width: 600px;">

            <h5 class="fw-bold text-primary text-
uppercase">Login</h5>

            <h1 class="mb-0">Please Login</h1>

        </div>

        <div class="row g-5">

            <div class="col-lg-3 col-md-3"></div>

            <div class="col-lg-6 col-md-6 wow zoomIn" data-wow-
delay="0.3s">

                <form class="singup-form" action="" method="post">

                    <?php

                        if(isset($errors)){

                            foreach ($errors as $error){ ?>

                                <div class="alert alert-danger"
style="width: 100%">

                                    <a href="#" class="close" data-
dismiss="alert" aria-label="close">&times;</a>

                                    <strong>Error!</strong> <?php echo
$error; ?>

                                </div>

                            <?php      }

```

```

    }
    ?>

    <div class="form-group mb-2">
        <label>Type</label>
        <select class="form-control bg-white"
name="type">
            <option value="">-- select type --
        </option>
            <?php foreach (get_types() as $key
=> $value ){ ?>
                <option value="<?php echo $key
?>" <?php if(isset($_POST['type'])){ if($_POST['type'] == $key){
echo 'selected'; } } ?> > <?php echo $value; ?> </option>
                <?php } ?>
            </select>
        </div>
        <div class="form-group mb-2">
            <label>Username</label>
            <input type="text" class="form-control"
name="username" value="<?php if(isset($_POST['username'])){ echo
$_POST['username']; } ?>">
        </div>
        <div class="form-group mb-2">
            <label>Password</label>
            <input type="password" class="form-
control" name="password">
        </div>
        <div class="">
            <div class="col-12">
                <p>
                    <a href="register_type.php"
class="text-info"> Create new account?</a>

```

```

        </p>

        <p>

        <a
href="forget_password.php"        class="text-info">        <a
Password?</a>        Forgot

        </p>

    </div>

</div>

    <div class="form-group mt-3">

        <button type="submit"
name="login" class="btn btn-primary">Login</button>

    </div>

</form>

</div>

</div>

</div>

</div>

</div>

<!-- Service End -->

<?php include 'theme/footer.php';?>

</body>

</html>

```

8.2 Register

```

<!DOCTYPE html>

<html lang="en">

<?php include "theme/header.php";

```

```

$title = 'Register';

if(!isset($_GET['type']) || empty($_GET['type'])){
    header('location:index.php');
    exit();
}

$type = trim($_GET['type']);

if(!in_array($type,['officer','donor'])){
    header('location:index.php');
    exit();
}

$table = $type;

if (isset($_POST['register'])) {
    $errors = array();

    if($table == 'donor') {
        $fields = array('Name' => 'name', 'E-mail' => 'email',
        'Birthdate' => 'birthdate', 'Diseases'=>'diseases', 'Blood
        Type'=>'blood_type', 'Mobile' => 'mobile', 'Username' =>
        'username', 'Password' => 'password', 'Confirm Password' =>
        'password_confirm');

        }elseif ($table == 'officer'){
            $fields = array('Name' => 'name', 'E-mail' => 'email',
            'Mobile' => 'mobile', 'Address' => 'address', 'Username' =>
            'username', 'Password' => 'password', 'Confirm Password' =>
            'password_confirm');
        }

        foreach ($fields as $key=> $field) {
            if (!isset($_POST[$field]) || empty(trim($_POST[$field])))
            {
                $errors[$field] = "$key field is required";
            }
        }
    }
}

```

```

        if($table == 'officer'){
            $address = mysqli_real_escape_string($conn,
trim($_POST['address']));
        }elseif ($table == 'donor'){
            $birthdate = mysqli_real_escape_string($conn,
trim($_POST['birthdate']));
            $diseases = mysqli_real_escape_string($conn,
trim($_POST['diseases']));
            $blood_type = mysqli_real_escape_string($conn,
trim($_POST['blood_type']));
        }

        $name = mysqli_real_escape_string($conn,
trim($_POST['name']));

        if (isset($_POST['mobile']) && !empty(trim($_POST['mobile'])))
        {
            $mobile = mysqli_real_escape_string($conn,
trim($_POST['mobile']));

            if (strlen($mobile) != 10) {
                $errors['mobile'] = 'Mobile number must contain 10
numbers';
            } elseif (!check_exist($mobile,"mobile",$table,'id')) {
                $errors['mobile'] = 'Mobile number already exists';
            } elseif (!check_number($mobile)) {
                $errors['mobile'] = 'Mobile number must contain numbers
only';
            }
        }
    }

    if (isset($_POST['email']) && !empty(trim($_POST['email']))) {
        $email = mysqli_real_escape_string($conn,
trim($_POST['email']));

        if (!check_exist($email,"email",$table,'id')) {

```

```

        $errors['email'] = 'E-mail already exists';
    } elseif (!valid_email($email)) {
        $errors['email'] = 'E-mail invalid';
    }
}

        if (isset($_POST['username']) &&
!empty(trim($_POST['username']))) {
            $username = mysqli_real_escape_string($conn,
trim($_POST['username']));

            if (!check_exist($username,"username",$table,'id')) {
                $errors['username'] = 'Username already exists';
            }
        }

$password = '';

        if (isset($_POST['password']) &&
!empty(trim($_POST['password']))) {
            $password = mysqli_real_escape_string($conn,
trim($_POST['password']));

            if (strlen($password) < 3) {
                $errors['password'] = 'Password must be at least 3
characters long';
                $password = '';
            }
        }

        $password_conf = mysqli_real_escape_string($conn,
trim($_POST['password_confirm']));

        if (!empty($password) && !empty($password_conf)) {
            if ($password != $password_conf) {
                $errors['password'] = 'Password does not match';
            }
        }
}

```

```

        if (empty($errors)) {
            if($table == 'officer') {
                $sql = "INSERT INTO `{$table}` (`email`, `name`, `mobile`,
`address`, `username`, `password`)
                . " VALUES
('$email','$name','$mobile','$address','$username','$password')";

            }elseif ($table == 'donor'){
                $sql = "INSERT INTO `{$table}` (`email`, `name`, `mobile`,
`birthdate`, `blood_type`, `diseases`, `username`, `password`)
                . " VALUES
('$email','$name','$mobile','$birthdate','$blood_type','$diseases
','$username','$password')";
            }

            if ($conn->query($sql) == TRUE) {
                header("location:thank2.php");
                exit();
            } else {
                echo $conn->error . '<br>';
                echo $sql;
            }
        }
    }
}
?>

```

```
<body>
```

```
<!-- Spinner Start -->
```

```
<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">
```

```
<div class="spinner"></div>
```

```

</div>

<!-- Spinner End -->


<!-- Topbar Start -->
<?php include "theme/topbar.php"; ?>
<!-- Topbar End -->


<!-- Navbar & Carousel Start -->
<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>
    <?php include 'theme/top_page.php'; ?>

</div>

<!-- Navbar & Carousel End -->


<!-- Service Start -->
<div class="container-fluid pb-5 wow fadeInUp" data-wow-
delay="0.1s">
    <div class="container">
        <div class="row g-5">
            <div class="col-lg-3 col-md-3"></div>
            <div class="col-lg-6 col-md-6 wow zoomIn" data-wow-
delay="0.3s">
                <form action="" method="post"
enctype="multipart/form-data">
                    <div class="row">
                        <?php
                            if(isset($errors)){
                                foreach ($errors as $error){ ?>

```



```

                                <div class="alert alert-danger"
style="width: 100%">

                                <a href="#" class="close" data-
dismiss="alert" aria-label="close">&times;</a>

                                <strong>Error!</strong> <?php
echo $error; ?>

                                </div>

                                <?php    }
                                }
                                ?>

                                <?php
                                if(isset($success)){
                                    ?>

                                <div class="alert alert-success"
style="width: 100%">

                                <a href="#" class="close" data-
dismiss="alert" aria-label="close">&times;</a>

                                <strong>Success!</strong> <?php
echo $success; ?>

                                </div>

                                <?php
                                }
                                ?>

                                <div class="form-group mb-3 col-md-12">

                                    <label>Name</label>

                                    <input type="text" maxlength="50"
class="form-control" name="name" value="<?php
if(isset($_POST['name'])) { echo $_POST['name']; } ?>">

                                </div>

                                <div class="col-md-6 mb-3">

                                    <label>E-mail</label>

```

```

                                <input type="email" maxlength="60"
class="form-control"          name="email"          value="<?php
if(isset($_POST['email']))){ echo $_POST['email']; } ?>">

                                </div>

                                <div class="col-md-6 mb-3">

                                    <label>Mobile</label>

                                <input type="text" maxlength="10"
class="form-control"          name="mobile"          value="<?php
if(isset($_POST['mobile']))){ echo $_POST['mobile']; } ?>">

                                </div>

                                <?php

                                if($type == 'officer'){

                                    ?>

                                <div class="form-group mb-3 col-md-12">

                                    <label>Address</label>

                                <input type="text" maxlength="200"
class="form-control"          name="address"          value="<?php
if(isset($_POST['address']))){ echo $_POST['address']; } ?>">

                                </div>

                                <?php }elseif ($type == 'donor'){

                                    ?>

                                <div class="col-md-6 mb-3">

                                    <label>Blood Type</label>

                                    <select class="form-control bg-
white" name="blood_type">

                                        <option value="">-- select
blood type --</option>

                                        <?php foreach (getBloods() as
$blood ){ ?>

                                            <option value="<?php echo
$blood; ?>" <?php if(isset($_POST['blood_type'])){
if($_POST['blood_type'] == $blood){ echo 'selected'; } } ?> > <?php
echo $blood; ?> </option>

```

```

        <?php } ?>

    </select>

</div>

<div class="col-md-6 mb-3">

    <label>Diseases</label>

    <textarea type="text" maxlength="200"
class="form-control" name="diseases"><?php
if(isset($_POST['diseases']))){ echo $_POST['diseases']; }
?></textarea>

</div>

<div class="col-md-6 mb-3">

    <label>Birthdate</label>

    <input type="date" class="form-
control" name="birthdate" value="<?php
if(isset($_POST['birthdate']))){ echo $_POST['birthdate']; } ?>">

</div>

<?php } ?>

<div class="col-md-6 mb-3">

    <label>Username</label>

    <input type="text" maxlength="15"
class="form-control" name="username" value="<?php
if(isset($_POST['username']))){ echo $_POST['username']; } ?>">

</div>

<div class="col-md-6 mb-3">

    <label>Password</label>

    <input type="password" class="form-
control" name="password">

</div>

<div class="col-md-6 mb-3">

    <label>Confirm Password</label>

```

```

                <input type="password"  maxlength="15"
class="form-control form-control-lg" name="password_confirm">

            </div>

            <div class="col-md-12">

                <button type="submit" name="register"
class="btn btn-primary">Register</button>

            </div>

        </div>

    </form>

</div>

</div>

</div>

</div>

<!-- Service End -->

<?php include 'theme/footer.php';?>

</body>

</html>

```

8.3 Add need

```

<!DOCTYPE html>

<html lang="en">

<?php include "theme/header.php";

if (!isset($_SESSION['user_id'])) {
    header('location:index.php');
}

```

```

        exit();
    }

    $title = 'Add Blood Need';
    $table = $_SESSION['table'];
    $id = $_SESSION['user_id'];
    if(isset($_POST['register'])) {
        $errors = array();

        $fields = array('Blood
Type'=>'blood_type', 'Amount'=>'amount', 'date'=>'datetime');

        foreach ($fields as $key=>$field){
            if(!isset($_POST[$field]) || empty(trim($_POST[$field]))){
                $errors[$field] = "$key field is required";
            }
        }
    }

    $blood_type =
mysqli_real_escape_string($conn,trim($_POST['blood_type']));

    if(isset($_POST['amount']) && !empty($_POST['amount'])){
        $amount =
mysqli_real_escape_string($conn,trim($_POST['amount']));

        if(!is_numeric($amount) || strlen($amount) > 6){
            $errors['amount'] = 'Invalid amount';
        }
    }

    if(isset($_POST['datetime']) && !empty($_POST['datetime'])){
        $datetime =
mysqli_real_escape_string($conn,trim($_POST['datetime']));

        $date = explode('T',$datetime);
        $dateNew = $date[0];
    }

```

```

        if(isset($date[1])){
            $datetime = $dateNew.' '.$date[1].':00';
        }
        if(!validateDate($datetime)){
            $errors['datetime'] = 'Invalid date ';
        }
    }

    if(empty($errors)){
        $sql = "INSERT INTO `need`
        (`blood_type`,`datetime`,`amount`,`officer_id`)
        VALUES
        ('$blood_type','$datetime','$amount','$id')";

        if ($conn->query($sql) == TRUE) {
            header('location:blood_need.php');
            exit();
        }else{
            echo $conn->error.'<br>';
            echo $sql;
        }
    }
}

?>

<body>

<!-- Spinner Start -->

<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">

    <div class="spinner"></div>

</div>

```

```

<!-- Spinner End -->

<!-- Topbar Start -->
<?php include "theme/topbar.php"; ?>
<!-- Topbar End -->

<!-- Navbar & Carousel Start -->
<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>
    <?php include 'theme/top_page.php'; ?>

</div>
<!-- Navbar & Carousel End -->

<!-- Service Start -->
<div class="container-fluid wow fadeInUp" data-wow-delay="0.1s">
    <div class="container">
        <div class="row g-5">
            <?php include 'theme/sidebar.php' ?>
            <div class="col-lg-9">
                <div class="mb-5">
                    <form class="singup-form" action=""
method="post" enctype="multipart/form-data">
                        <div class="row">
                            <?php
                                if(isset($errors)){
                                    foreach ($errors as $error){ ?>

```

```

        <div class="alert alert-danger"
style="width: 100%">

            <a href="#" class="close"
data-dismiss="alert" aria-label="close">&times;</a>

            <strong>Error!</strong>

<?php echo $error; ?>

        </div>

<?php    }

}

?>

<div class="col-md-12 mb-3">

    <label>Blood Type</label>

    <select class="form-control bg-
white" name="blood_type">

        <option value="">-- select
blood type --</option>

        <?php foreach (getBloods() as
$blood ){ ?>

            <option value="<?php echo
$blood;    ?>"    <?php    if(isset($_POST['blood_type']))){
if($_POST['blood_type'] == $blood){ echo 'selected'; } } ?> > <?php
echo $blood; ?> </option>

            <?php } ?>

        </select>

    </div>

    <div class="col-md-6 mb-3">

        <label>Amount</label>

        <input type="text" maxlength="6"
name="amount"        class="form-control"        value="<?php    if
(isset($_POST['amount']))){echo $_POST['amount']; } ?>" >

    </div>

    <div class="col-md-6 mb-3">

```



```

        <label>Date</label>

        <input type="datetime-local"
name="datetime"      class="form-control"      value="<?php      if
(isset($_POST['datetime']))){echo $_POST['datetime']; } ?>" >

    </div>

    <div class="col-md-12">

        <button type="submit" name="register"
class="btn btn-primary">Add</button>

    </div>

</div>

</form>

</div>

</div>

</div>

</div>

</div>

<!-- Service End -->

<?php include 'theme/footer.php';?>

</body>

</html>

```

8.4 Reserve

```

<!DOCTYPE html>

<html lang="en">

<?php include "theme/header.php";

$title = 'Reserve appointment';

```

```

$form = 1;

if(isset($_GET['form']) && !empty($_GET['form'])){
    $form = $_GET['form'];
}

$question = questions('s'.$form);

if (!$question) {
    header('location:list.php');
    exit();
}

$formInput = 's'.$form;

if(isset($_POST[$formInput])){
    $field = 'r'.$form;

    if ($question) {
        if (!isset($_POST[$field]) || empty(trim($_POST[$field])))
        {
            $errors[$field] = "Please answer";
        }

        if (empty($errors)) {
            $a = mysqli_real_escape_string($conn,
trim($_POST[$field]));

            if ($a == $question['next']) {
                $form++;

                if (isset($question['end'])){
                    header("location:list.php?");
                    exit();
                }

                header("location:reserve.php?form=$form");
                exit();
            }
        }
    }
}

```

```

        }

        header("location:sorry.php");

        exit();

    }

}

?>

<body>

<!-- Spinner Start -->

<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">

    <div class="spinner"></div>

</div>

<!-- Spinner End -->

<!-- Topbar Start -->

<?php include "theme/topbar.php"; ?>

<!-- Topbar End -->

<!-- Navbar & Carousel Start -->

<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>

    <?php include 'theme/top_page.php'; ?>

</div>

<!-- Navbar & Carousel End -->

```

```

<!-- Service Start -->

<div class="container-fluid pb-5 wow fadeInUp" data-wow-
delay="0.1s">

    <div class="container">

        <div class="section-title text-center position-relative pb-
3 mb-5 mx-auto" style="max-width: 600px;">

            <h5 class="fw-bold text-primary text-uppercase"><?php
echo $title; ?></h5>

            <h1 class="mb-0">Please fill out the following
questionnaire / <?php echo $form.'-'.count(getQuestion()); ?></h1>

        </div>

        <div class="row g-5">

            <div class="col-lg-3 col-md-3"></div>

            <div class="col-lg-6 col-md-6 wow zoomIn" data-wow-
delay="0.3s">

                <?php

                if(isset($errors)){

                    foreach ($errors as $error){ ?>

                        <div class="alert alert-danger" style="width:
100%">

                            <a href="#" class="close" data-
dismiss="alert" aria-label="close">&times;</a>

                            <strong>Error!</strong> <?php echo
$error; ?>

                        </div>

                    <?php    }

                }

                ?>

                <form method="post">

```

```

        <p><?php echo $form.'- '.$question['question'];
?></p>

        <div class="form-check">

            <input class="form-check-input" type="radio"
name="r<?php echo $form ?>" id="exampleRadios1" value="true">

                <label class="form-check-label"
for="exampleRadios1">

                    Yes

                </label>

            </div>

            <div class="form-check">

                <input class="form-check-input" type="radio"
name="r<?php echo $form ?>" id="exampleRadios2" value="false">

                    <label class="form-check-label"
for="exampleRadios2">

                        No

                    </label>

                </div>

            <div class="form-group mt-3">

                <button type="submit" name="s<?php echo
$form ?>" class="btn btn-primary"> <i class="bi bi-arrow-
right"></i></button>

            </div>

        </form>

    </div>

</div>

</div>

</div>

</div>

<!-- Service End -->

```

```

<?php include 'theme/footer.php';?>

</body>

</html>

```

8.5 View reservations

```

<!DOCTYPE html>

<html lang="en">

<?php include "theme/header.php";

$title = 'Manage Reservation';
$table = $_SESSION['table'];
$id = $_SESSION['user_id'];

if (!isset($_SESSION['user_id'])) {
    header('location:index.php');
    exit();
}

?>

<body>

<!-- Spinner Start -->

<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">

    <div class="spinner"></div>

</div>

<!-- Spinner End -->

```

```

<!-- Topbar Start -->

<?php include "theme/topbar.php"; ?>

<!-- Topbar End -->


<!-- Navbar & Carousel Start -->

<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>

    <?php include 'theme/top_page.php'; ?>

</div>

<!-- Navbar & Carousel End -->


<!-- Service Start -->

<div class="container-fluid wow fadeInUp" data-wow-delay="0.1s">
    <div class="container">
        <div class="row g-5">
            <?php include 'theme/sidebar.php' ?>
            <div class="col-lg-9">
                <div class="mb-5">
                    <table class="table table-hover">
                        <thead>
                            <tr>
                                <th scope="col">Date / Time</th>
                                <th scope="col">Donor</th>
                                <th scope="col">Donor Blood</th>
                                <th scope="col">Mobile</th>
                                <th scope="col">Status</th>
                                <th scope="col">Action</th>

```

```

        </tr>

    </thead>

    <tbody>

    <?php

        $sql = "select * from `reservation` where
officer_id = $id ";

        $result = $conn->query($sql);

        if($result->num_rows > 0) {
            while($row = $result->fetch_assoc()) {
                $donor = getDonor($row['donor_id']);

                ?>

                <tbody>

                <tr>

                    <td><?php echo $row['date'].' /
'.$row['time']; ?></td>

                    <td><?php echo $donor['name']; ?></td>

                    <td><?php echo $donor['blood_type'];
?></td>

                    <td><?php echo $donor['mobile']; ?></td>

                    <td><strong class="<?php echo
status($row['status'])['color']; ?>"><?php
status($row['status'])['name']; ?></strong></td>

                    <td>

                        <?php if($row['status'] == 0){ ?>

                            <a href="o_status.php?id=<?php
echo $row['id'] ?>&status=2" class="text-success" onclick="return
confirm('Are you sure approved?')"> Approved </a>

                            <?php } ?>

                        </td>

                    </tr>

                </tbody>

            <?php }} ?>

```



```

        </table>
    </div>
</div>
</div>
</div>
</div>
<!-- Service End -->

<?php include 'theme/footer.php';?>
</body>

</html>

```

8.6 Update profile

```

<!DOCTYPE html>
<html lang="en">
<?php include "theme/header.php";
$title = 'Profile';
$table = $_SESSION['table'];
$id = $_SESSION['user_id'];

if (!isset($_SESSION['user_id'])) {
    header('location:index.php');
    exit();
}

if (isset($_POST['register'])) {
    $errors = array();

```

```

        if($table == 'donor') {

            $fields = array('Name' => 'name', 'E-mail' => 'email',
'Birthdate'      =>      'birthdate','Diseases'=>'diseases','Blood
Type'=>'blood_type',    'Mobile'    =>    'mobile',    'Username'    =>
'username');

        }elseif ($table == 'officer'){

            $fields = array('Name' => 'name', 'E-mail' => 'email',
'Mobile' => 'mobile', 'Address' => 'address', 'Username' =>
'username');

        }else{

            $fields = array('Name' => 'name', 'E-mail' => 'email',
'Mobile' => 'mobile', 'Username' => 'username');

        }

        foreach ($fields as $key=> $field) {

            if (!isset($_POST[$field]) || empty(trim($_POST[$field])))
{

                $errors[$field] = "$key field is required";

            }

        }

        if($table == 'officer'){

            $address    =    mysqli_real_escape_string($conn,
trim($_POST['address']));

        }elseif ($table == 'donor'){

            $birthdate    =    mysqli_real_escape_string($conn,
trim($_POST['birthdate']));

            $diseases    =    mysqli_real_escape_string($conn,
trim($_POST['diseases']));

            $blood_type    =    mysqli_real_escape_string($conn,
trim($_POST['blood_type']));

        }

        $name    =    mysqli_real_escape_string($conn,
trim($_POST['name']));

```

```

        if (isset($_POST['mobile']) && !empty(trim($_POST['mobile'])))
        {
            $mobile = mysqli_real_escape_string($conn,
            trim($_POST['mobile']));

            if (strlen($mobile) != 10) {

                $errors['mobile'] = 'Mobile number must contain 10
numbers';

            } elseif (!check_exist($mobile,"mobile",$table,'id',$id))
            {

                $errors['mobile'] = 'Mobile number already exists';

            } elseif (!check_number($mobile)) {

                $errors['mobile'] = 'Mobile number must contain numbers
only';

            }

        }

        if (isset($_POST['email']) && !empty(trim($_POST['email']))) {

            $email = mysqli_real_escape_string($conn,
            trim($_POST['email']));

            if (!check_exist($email,"email",$table,'id',$id)) {

                $errors['email'] = 'E-mail already exists';

            } elseif (!valid_email($email)) {

                $errors['email'] = 'E-mail invalid';

            }

        }

        if (isset($_POST['username']) &&
!empty(trim($_POST['username']))) {

            $username = mysqli_real_escape_string($conn,
            trim($_POST['username']));

            if (!check_exist($username,"username",$table,'id',$id)) {

                $errors['username'] = 'Username already exists';

            }

        }
    }

```

```

        if (empty($errors)) {

            if($table == 'officer') {

                $sql = "UPDATE `{$table}` SET `email`='$email',
`name`='$name', `mobile`='$mobile', `address`='$address',
`username`='$username' where id = $id";

                }elseif ($table == 'donor'){

                    $sql = "UPDATE `{$table}` SET `email`='$email',
`name`='$name', `mobile`='$mobile',
`birthdate`='$birthdate', `diseases`='$diseases', `blood_type`='$blood_type', `username`='$username' where id = $id";

                }elseif ($table == 'admin'){

                    $sql = "UPDATE `{$table}` SET `email`='$email',
`name`='$name', `mobile`='$mobile', `username`='$username' where
id = $id";

                }

                if ($conn->query($sql) == TRUE) {

                    $success = 'Update successfully';

                } else {

                    echo $conn->error . '<br>';

                    echo $sql;

                }

            }

        }

    ?>

```

```

<body>

```

```

<!-- Spinner Start -->

```

```

<div id="spinner" class="show bg-white position-fixed translate-
middle w-100 vh-100 top-50 start-50 d-flex align-items-center
justify-content-center">

```

```

    <div class="spinner"></div>

```

```

</div>

<!-- Spinner End -->

<!-- Topbar Start -->
<?php include "theme/topbar.php"; ?>
<!-- Topbar End -->

<!-- Navbar & Carousel Start -->
<div class="container-fluid position-relative p-0">

    <?php include "theme/menu.php"; ?>
    <?php include 'theme/top_page.php'; ?>

</div>

<!-- Navbar & Carousel End -->

<!-- Service Start -->
<div class="container-fluid wow fadeInUp" data-wow-delay="0.1s">
    <div class="container">
        <div class="row g-5">
            <?php include 'theme/sidebar.php' ?>
            <div class="col-lg-9">
                <div class="mb-5">
                    <?php
                    $sql = "select * from `\$table` where `id`='\$id'
limit 1";

                    $result = $conn->query($sql);
                    if($result->num_rows > 0) {
                        while($row = $result->fetch_assoc()) {

```

```

        ?>

        <form action="" method="post"
enctype="multipart/form-data">

        <div class="row">

            <?php
                if(isset($errors)){
                    foreach ($errors as $error){
                        ?>

                            <div class="alert alert-
danger" style="width: 100%">

                                <a href="#"
class="close" data-dismiss="alert" aria-label="close">&times;</a>

                                    <strong>Error!</strong>
                                <?php echo $error; ?>

                            </div>

                        <?php    }
                    }
                ?>

            <?php
                if(isset($success)){
                    ?>

                        <div class="alert alert-
success" style="width: 100%">

                            <a href="#" class="close"
data-dismiss="alert" aria-label="close">&times;</a>

                                <strong>Success!</strong>
                            <?php echo $success; ?>

                        </div>

                        <?php
                    }
                ?>

```

```

col-md-12">
    <div class="form-group mb-3
        <label>Name</label>
        <input type="text"
maxlength="50" class="form-control" name="name" value="<?php
if(isset($_POST['name']))){ echo $_POST['name']; }else{ echo
$row['name']; } ?>">
    </div>
    <div class="col-md-6 mb-3">
        <label>E-mail</label>
        <input
type="email" maxlength="60" class="form-control" name="email"
value="<?php if(isset($_POST['email']))){ echo $_POST['email'];
}else{ echo $row['email']; } ?>">
    </div>
    <div class="col-md-6 mb-3">
        <label>Mobile</label>
        <input type="text"
maxlength="10" class="form-control" name="mobile" value="<?php
if(isset($_POST['mobile']))){ echo $_POST['mobile']; }else{ echo
$row['mobile']; } ?>">
    </div>
    <?php
if($table == 'officer'){
    ?>
    <div class="form-group mb-
3 col-md-12">
        <label>Address</label>
        <input type="text"
maxlength="200" class="form-control" name="address" value="<?php
if(isset($_POST['address']))){ echo $_POST['address']; }else{ echo
$row['address']; } ?>">
    </div>
    <?php }elseif ($table ==
'donor'){

```

```

        ?>

        <div class="col-md-6 mb-
3">

        <label>Blood Type</label>

        <select class="form-
control bg-white" name="blood_type">

        <option value="">-
- select blood type --</option>

        <?php foreach
(getBloods() as $blood ){ ?>

        <option
value="<?php echo $blood; ?>" <?php
if(isset($_POST['blood_type'])){ if($_POST['blood_type'] ==
$blood){ echo 'selected'; } }else{ if($row['blood_type'] ==
$blood){ echo 'selected'; } } ?> > <?php echo $blood; ?> </option>

        <?php } ?>

        </select>

        </div>

        <div class="col-md-6 mb-
3">

        <label>Diseases</label>

        <textarea type="text"
maxlength="200" class="form-control " name="diseases"><?php
if(isset($_POST['diseases'])){ echo $_POST['diseases']; }else{
echo $row['diseases']; } ?></textarea>

        </div>

        <div class="col-md-6 mb-
3">

        <label>Birthdate</label>

        <input type="date"
class="form-control " name="birthdate" value="<?php
if(isset($_POST['birthdate'])){ echo $_POST['birthdate']; }else{
echo $row['birthdate']; } ?>">

```



```

        </div>

        <?php } ?>

        <div class="col-md-6 mb-3">

            <label>Username</label>

            <input type="text"
maxlength="15" class="form-control" name="username" value="<?php
if(isset($_POST['username'])) { echo $_POST['username']; } else {
echo $_row['username']; } ?>">

        </div>

        <div class="col-md-12">

            <button type="submit"
name="register" class="btn btn-primary">Update</button>

        </div>

    </div>

</form>

<?php }} ?>

</div>

</div>

</div>

</div>

<!-- Service End -->

<?php include 'theme/footer.php';?>

</body>

</html>

```